

Workload Distribution

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Preliminary Stage II Report

Gearbox Design Project

MECH 360 - Design of Mechanical Elements
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Abstract

In this initial design report, we outline the development of a gearbox intended for a 30 MW Concentrated Solar Power plant, part of the Crescent Dunes Energy Project by the ACS. Our goal is to engineer an efficient and reliable gearbox that links a high capacity steam turbine to a synchronous generator. This project aims to address the rising demand for sustainable energy. Our current design submission focuses on the mechanical configuration, load capacities, initial shaft design, gear sizing and selection, and bearing selection. This will act as a reference for the final analysis on the fatigue effects of the gearbox, to take place in future analysis after feedback from the administrative team.

Table of Contents

1.0 Introduction & Background.....	6
1.1 Introduction.....	6
1.2 CSP System Overview.....	6
2.0 Review of Literature.....	7
4.0 Design Objectives & Requirements.....	9
5.0 Concept Design - Overall.....	10
6.0 Concept Design - Gearbox Elements.....	11
6.1 Housing Design.....	11
6.2 Material Choice.....	11
7.0 Concept Design - Speeds and Loads.....	12
7.1 Shaft Speeds.....	12
7.2 Shaft Loads.....	12
8.0 Gearbox Element Design.....	15
8.1 Gearbox Layouts, Drawings & Cross-sections.....	15
8.2 Preliminary Sizing & Specifications.....	18
8.2.1 Gears.....	19
8.2.2 Bearings.....	20
8.2.3 Couplings.....	21
8.2.4 Lock Nut and Clip.....	22
8.2.5 Seals.....	22
8.2.6 Housing.....	22
8.3 Potentially Critical Points.....	23
9.0 Preliminary Design for Shafts.....	24
9.1 Shaft 1 Calculations.....	25
9.2 Shaft 2 Calculations.....	25
9.3 Shaft 3 Calculations.....	27
9.4 Shaft 4 Calculations.....	28
10.0 Assembly Process.....	29
11.0 Concluding Remarks and Next Steps.....	29
References.....	31
Appendix A. Bearing Data Sheets and Calculations.....	33
Appendix B. Gear Stress Calculation Sheets.....	40
Appendix C. Shaft Stress Calculation Sheets.....	43

List of Figures

Figure 1. Simplified diagram of the proposed CSP System.....	5
Figure 2. The gearbox analyzed for redesign by the GRC, as typical in wind turbines.....	6
Figure 3. Schematic diagram of the drivetrain of 5MW wind turbine.....	7
Figure 4. Concept design with gear ratios.....	9
Figure 5. Front and back view of gearbox housing.....	10
Figure 6. Daily Power Generation Profile.....	13
Figure 7. Torque Load on Gearbox Shafts as a Result of Input Torque on a 24 Hour Cycle.....	13
Figure 8. Gear-box Cross-Section.....	15
Figure 9. Shaft 1 Cross Section.....	15
Figure 10. Shaft 2 Cross Section.....	16
Figure 11. Shaft 3 Cross Section.....	16
Figure 12. Shaft 4 Cross Section.....	17
Figure 13. Gear and Bearing Naming and Locations.....	17
Figure 14. Bearing Selection from Calculations.....	20
Figure 15. Potential Critical Locations.....	22
Figure 16: Gear Box Assembly, cylinders represent pitch diametre and face width.....	23
Figure 17. Shaft 1 FBD.....	25
Figure 18. Shaft 2 FBD.....	26
Figure 19. Shaft 3 FBD.....	27
Figure 20. Shaft 4 FBD.....	28

List of Tables

Table 1. Gearbox Objectives with justification.....	8
Table 2. Turbine-Gearbox generator System Characteristics.....	13
Table 3. Varying Gear Speeds due to Electrical Generation Profile and Output Generation due to Efficiencies.....	14
Table 4. Gear Specification Summary.....	18
Table 5. Lock nut selection for fixed bearings.....	21
Table 6. Seal selection.....	21
Table 7. Process to Assemble Gearbox.....	29

1.0 Introduction & Background

1.1 Introduction

This technical report launches the design process for a gearbox to be used in a 30 MW Concentrated Solar Power (CSP) plant, specifically designed for the Crescent Dunes Solar Energy Project managed by ACS. The goal is to engineer a gearbox that effectively couples a high-capacity steam turbine to a synchronous generator, optimizing the conversion of thermal energy into electrical power. This gearbox is essential for maximizing the efficiency of power transmission and ensuring operational reliability under varying load conditions.

The context of this project is around the increasing demand for renewable energy solutions that are both sustainable and capable of integrating into existing power grids. The design challenges include maintaining high efficiency, minimizing wear and tear, and ensuring compatibility with the demands of solar power generation. This stage 2 report dives deep into the load capacities, mechanical configurations, gear sizing, shaft design, and bearing selection.

1.2 CSP System Overview

The proposed Concentrated Solar Power (CSP) System involves an array of thousands of mirrors all reflecting sunlight towards a receiving tower positioned in the center. The sunlight heats up the chamber on the tower to temperatures of 580°C . This chamber contains a salt mixture of 60% sodium nitrate and 40% potassium nitrate that becomes molten due to the intense heat. Once the salt becomes molten, it is moved to a thermal storage unit that feeds a steam generator once necessary. The steam powers the turbine that is connected to a generator via our proposed speed-reducing gearbox. The, now cooled, molten salt is finally transferred to the cool side thermal storage ready to be used again in the cycle as shown in Figure 1.

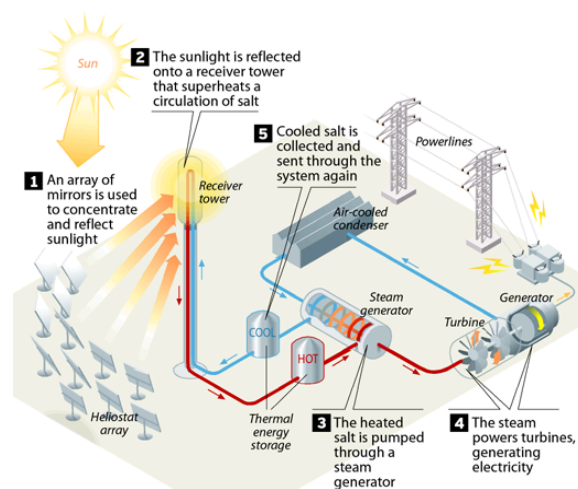


Figure 1. Simplified diagram of the proposed CSP System [3]

2.0 Review of Literature

In the 2011 report *Gearbox Reliability Collaborative Project Report: Findings from Phase 1 and Phase 2 Testing*, H. Link, et. al. examine common reliability shortcomings in wind turbine gearboxes [1]. The project, formally referred to as the Gearbox Reliability Collaborative (GRC) project, began taking form in 2006 with 5 objectives, centered on contributing to increased reliability in the gearbox design through the investigation of previous failures, current and future system loading, and advance field testing and monitoring techniques. Significant to this report, the GRC analyzed a common existing gearbox design implemented in wind turbines and offered several points of redesign to achieve increased reliability. The selected gearbox is shown in Figure 2 and features three stages to achieve an overall gear ratio of 1:81.491.

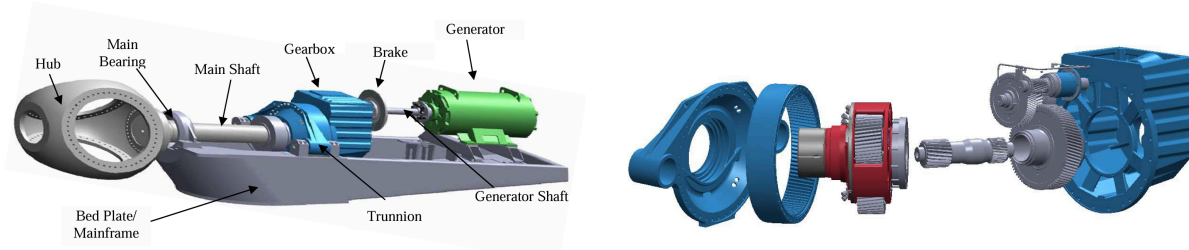


Figure 2. The gearbox analyzed for redesign by the GRC, as typical in wind turbines. Adapted from [1]

A significant point of redesign pertained to bearing selection on the high-speed shaft, low-speed shaft, and planetary gear system. The spherical roller bearings originally on the planet gears were found to have significantly reduced operating life. These bearings were replaced with cylindrical ones that were better suited to the application. Additionally, the spherical roller bearing located on the high-speed shaft reacted poorly to thrust and was recommended to be replaced with a tapered roller bearing. The GRC project report also presented findings regarding various effects of loading on system wear. The GRC explored non-torque loading and bending on the shafts, which were found to affect tooth contact patterns in the low-speed stage, resulting in a shorter gearbox life. This offers evidence that these loading types are not negligible, contradictory to what many experts previously believed. The findings of phases 1 and 2 of the GRC project demonstrate the necessity to analyze all loading cases in gearbox design and avoid focusing solely on the primary torque loads. Additionally, it shows evidence that cylindrical roller bearings should be selected for this project.

The 2015 report *Assessment of Gearbox Operational Loads and Reliability under High Mean Wind Speeds*, D. Dabrowski and A. Natarajan, Examined the loads and design of a gearbox [2]. This article looks at the dynamic loads occurring in the gearbox of a 5MW offshore wind turbine under high wind loading. The increased wind speed analyzed was 30m/s, creating higher stresses as well as stress under emergency stops. The design and analysis was based on an upscaled version of a 750 kW prototype for which physical tests were completed. The gearbox consisted of three stages, with an overall gear ratio of 1: 81.5. The use of a storm controller to limit the speed seen by the drivetrain from raised wind speeds was investigated.

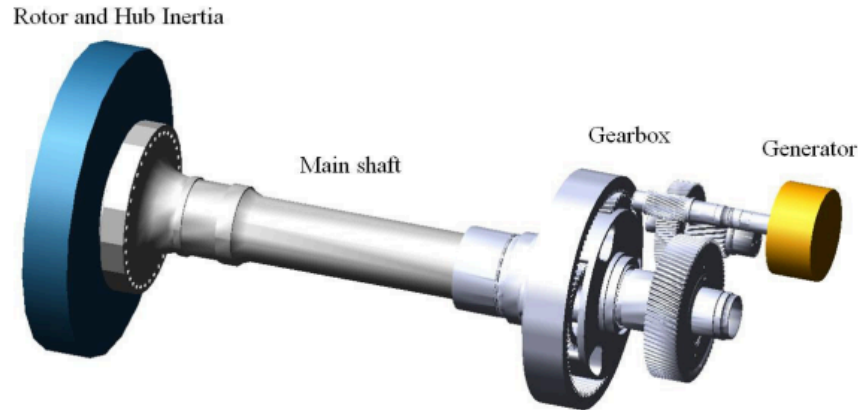


Figure 3. Schematic diagram of the drivetrain of 5MW wind turbine [2]

First, the loads were simulated for the gearbox for emergency shutdowns when wind speeds were too high. This was also repeated for large wind turbulence, introducing dynamic loading on the drivetrain. It was found that the highest forces seen in gear teeth were seen during the high wind and not during the shutdown operation. For this reason, it can be concluded that the use of emergency shutdowns do not have a significant influence on the ultimate loads in the gearbox. The reliability was also found to be high, having a value of 8.06×10^{-4} and reducing to only 2.75×10^{-5} when storm controller reduction was implemented. Overall, the lifetime of the drivetrain can be confidently extended by conducting regular shutdowns when operating conditions are outside of the design range without having an impact on the structure of the gears.

3.0 Market Survey

Our engineering team was hired by ACS [10], the owners of the Crescent Dunes Solar Energy Project in Nevada. They have given our team the bid to design a gearbox for a new smaller CSP plant with a 30 MW capacity. The plant will run from 5 pm to 5 am powering the electrical grid. Our goal is to create a gearbox with an efficiency of 96-97 percent. ACS in the past has built three other CSPs [10]. One of them was designed to output 100 MW, which in turn can power 75,000 homes [10]. The current design of 30 MW will power up to 22,500 homes. .

In an article written in 2024, *Global CSP installed cost 2010-2023*, Lucia Fernandez, The average cost per kilowatt installed over the past two years for CSP has been \$5510 USD [4]. The plant being designed currently is 30 MW, which will have an estimated cost of 165.3 million dollars USD (this estimate is not an exact amount and was done for an idea of the cost). In previous projects, the United States Government backed the 110 MW project with a 737 million dollar loan [10]. Since a project of a larger magnitude had funding, the 30 MW project is hopeful to acquire funding as well.

Nevada's energy comes from natural gas-fired plants that account for 57 percent of all in-state power [9]. The state of Nevada does not have large natural gas producers, therefore it is imported from other states and countries that charge a premium [9]. This cost is passed onto the population of Nevada resulting in an increase of cost of living. Building a 30 MW would decrease the cost of electricity to 22,500 homes by creating power in-state, as well as the plant releasing 0 tons of carbon in the environment [10].

4.0 Design Objectives & Requirements

Table 1 presents several requirements and objectives for the gearbox, each paired with a target metric (if applicable) and justification. Note that requirements are criteria that must be met for the design to be deemed acceptable. Objectives, while they should be treated as necessary criteria, have inherent flexibility. Objectives are used throughout this project to guide sensible design decisions.

Table 1. *Gearbox Objectives with justification*

Objective	Target	Justification
Input Shaft Speed (<i>requirement</i>)	6000 RPM	The turbines have a maximum output shaft speed of 6000 RPM. To avoid unnecessary losses and achieve maximal output, this must be matched.
Output Shaft Speed (<i>requirement</i>)	450 RPM	The generators are rated for a 450 RPM input. To achieve maximal generation under peak operating conditions, the gearbox output must match this speed.
Generator Power Output (<i>Requirement</i>)	15 MW	The gearbox must be designed such that the maximum power output of the generator is 15 MW.
In-plane Input and Output Shafts (<i>Requirement</i>)	Horizontal shaft alignment within 10mm	The input and output shafts are required to be at the same height above the steel support frame.
Suitable Lifetime (<i>requirement</i>)	25 Years	Externally provided requirement.
Mounting Points (<i>Requirement</i>)	Several external mounting points	The housing must be designed such that it can securely mount the steel supporting frame.
Integrated Lube Ports (<i>Requirement</i>)	At least 1 input and 1 output port	Lubrication is critical to extend the lifetime of the mechanical elements within the gearbox. Input and output lube ports must be included to facilitate this.
Fatigue Resistance (<i>Objective</i>)	Lifetime of all parts should exceed 5 years.	The gearbox must be capable of withstanding 24 hour cyclical loading. The cycle must be incorporated into load calculations to avoid premature part failure.
Easy serviceability (<i>Objective</i>)	Any one part is replaceable in under 8 hours.	The gearbox should be designed with ease of access in mind to facilitate maintenance. An emphasis should be placed on housing removal, bearing replacement, shaft replacement and gear replacement.
Safety Factor (<i>Objective</i>)	SF between 1.5 and 2.5	A reasonable safety factor is needed to minimize chance of premature failure while ensuring the gearbox is not over designed to a fault.

5.0 Concept Design - Overall

The layout seen below was decided upon for the gearbox. The gearbox consists of three stages. The first stage consists of the input from the steam turbine, where the power will enter the gearbox. This stage has a gear reduction of 3:1, reducing our input 6000 rpm to 2000 rpm entering stage 2. Stage two takes the 2000 rpm entering speed, and using a 20:9 gear ratio, reduces this to a value of 900 rpm. The final stage has a gear ratio of 2:1 which reduces our speed to the final output of 450 rpm. We decided on a three stage design as this would allow the gears not to be over sized but also not create an unnecessary number of gears and shafts. Each of the bearing locations have been chosen, allowing ample room for the necessary bearings and supports. The higher gear ratios were placed first intentionally, where the torque will be lower and allow for smaller gears, creating a smaller overall gearbox.

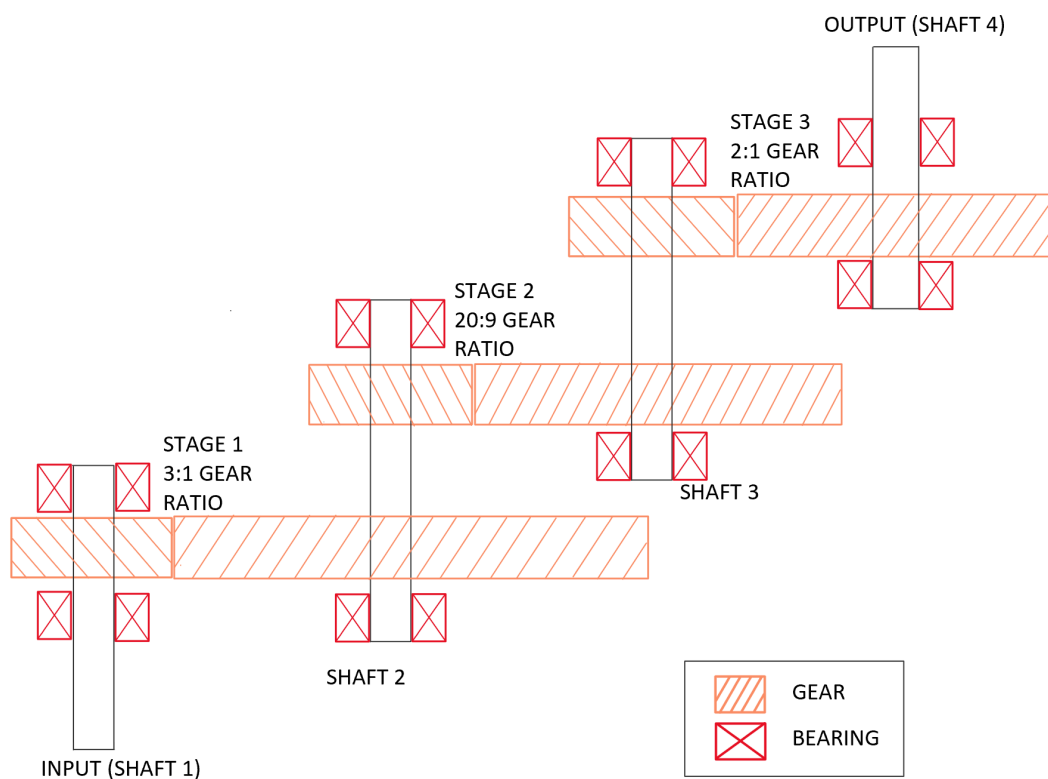


Figure 4. Concept design with gear ratios

6.0 Concept Design - Gearbox Elements

6.1 Housing Design

The housing serves the purpose of protecting the internal system from the environment as well as being the only supporting element for the shafts. The internal cavity of the housing will also contain the

lubricating oil. On the exterior of the housing, there will be cutouts on either side in order for the bearings and their respective shafts to be fixed as seen in Figure 5.

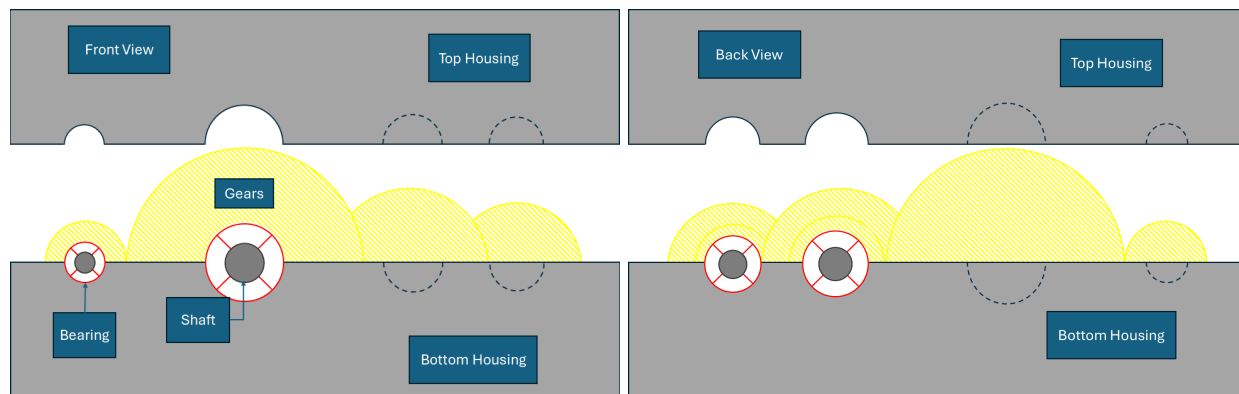


Figure 5. *Front and back view of gearbox housing*

The bottom and top housings will have extruded flanges that will allow us to bolt down the housing and to each other. The shafts will have bearings on either end, as shown in Figure 4. Two support columns will be located where the middle left bearing and the middle right bearing are located in Figure 4 with similar cutouts to those found on the exterior. Those bearings, along with the shafts, will be able to slide into the cutouts allowing the gear teeth to mesh.

6.2 Material Choice

Seeing as how our gears and shafts are going to be under large loads, we chose to use Nitrided AISI 4140 for our gears, and quench and tempered AISI 4140 for our shafts. This grade of steel offers robust strength, excellent toughness, and high wear resistance making it ideal for high-load and speed applications. [8] For our housing, we opted to go for a cast steel as it offers good strength, toughness, and allows for us to mold the more intricate portions of the housing.

7.0 Concept Design - Speeds and Loads

7.1 Shaft Speeds

The speed of the input (high-speed) shaft is required to be 6000 RPM at peak operation. By applying and rearranging standard gear equations for speed ratios, the corresponding speeds of the intermediate (mid-speed) shafts and output (low-speed) shaft are calculated. The following sample equation shows the expression used:

$$|e| = \left| \frac{\omega_2}{\omega_1} \right| = \frac{r_1}{r_2} \rightarrow \omega_2 = \omega_1 \frac{r_1}{r_2}$$

In the above equation, e is the speed ratio, ω is the shaft speeds in rad/s, and r is the radius of each gear. Note that the absolute value is removed from the final expression. This is possible as it is noted that the second shaft will rotate in the opposite direction of the first, and thus will always have the opposite sign. Additionally, by multiplying the speeds specified in RPM by 2π and dividing by 60, they are converted to rad/s. The speed of each shaft is summarized in Table 2.

7.2 Shaft Loads

To calculate the torque load that each shaft experiences, it is important to consider the varying input factors and efficiencies. To account for these factors, we need to find a function that will output the specific torque based on the power generated at a given instant. This is done using the relationships of power, torque, rotational speeds, and gear radius.

The gearbox is estimated to be 96% efficient, which aligns with typical transmission gear efficiencies, which range from 90% to 99% [6][7]. Taking this to be the average loss across the three stage gearbox, we have a loss between gear A and B, gear C and D, and gear E and F (as shown in Figure 4) of $\sqrt[3]{0.96}$.

The power output requirement of the gearbox is obtained by dividing the power output requirement of the generator by the generator's efficiency:

$$P_{GB,Out} = \frac{P_{Gen,Out}}{\eta_{Gen}} = \frac{P_{Gen,Out}}{0.96}$$

Applying the same concept to the power transmission through the gearbox, the power input that is actually required by the turbine is given as:

$$P_{GB,In} = \frac{P_{GB,Out}}{\eta_{GB}} = \frac{P_{Gen,Out}}{(0.96)^2}$$

where the $P_{Gen,Out}$ is read from the Daily Electrical Generation chart.

The force acting on a gear is given as,

$$F = \frac{P}{v} = \frac{P}{\omega \times r}$$

and torque as for the first gear as,

$$T_A = F_A \times r_A$$

Combining these two equations we have,

$$T_A = \left(\frac{P}{\omega_A} \right)$$

We can get a function for the torque of gear A that depends on the input power,

$$T_A = \frac{P_x}{\omega_A \left(\frac{2\pi}{60} \right)} [\text{Nm}]$$

Now that we have our torque and speed of gear A, we can now calculate the torque and speed of gear B, but we need to consider the efficiencies,

$$T_B = \frac{\sqrt[3]{\eta} P_x}{(\omega_B)} [\text{Nm}]$$

and subsequently,

$$\omega_B = \omega_A \frac{r_A}{r_B} [\text{rad/s}]$$

Since our gear B and C are connected on the same shaft, the $T_B = T_C$ and the $\omega_B = \omega_C$. Then for gear D we have,

$$T_D = \frac{\sqrt[3]{\eta} T_C \omega_C}{\omega_C} = \frac{\sqrt{\eta} T_C \omega_B}{\omega_D}$$

Since,

$$\omega_D = \frac{r_C \omega_B}{r_D} [\text{rad/s}]$$

then we have,

$$T_D = \frac{\sqrt[3]{\eta} r_D T_B}{r_C} [\text{Nm}]$$

Since our gear D and E are connected on the same shaft, the $T_D = T_E$ and the $\omega_D = \omega_E$. Then for gear F we have,

$$T_F = \frac{\sqrt[3]{\eta} T_E \omega_E}{\omega_E} = \frac{\sqrt{\eta} T_E \omega_D}{\omega_F}$$

Since,

$$\omega_F = \frac{r_E \omega_D}{r_F} [\text{rad/s}]$$

then we have,

$$T_F = \frac{\sqrt[3]{\eta} r_F T_D}{r_E} [\text{Nm}]$$

Now that we have our formulas to calculate ω and T , we have created the table below (Table 3) in excel using the functions and calculating the speeds and torques from hour 1 to 24.

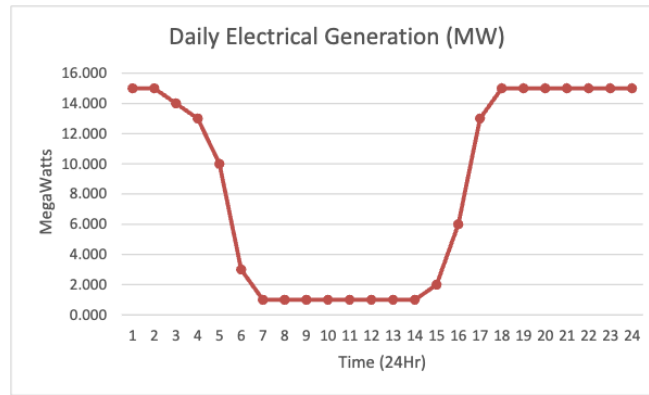


Figure 6. Daily Power Generation Profile

Table 2. Turbine-Gearbox generator System Characteristics

Item	Value	Units
Nominal rated electrical output power (per turbine)	15.0	MW
Assumed gearbox efficiency	96-97	%
Generator efficiency	96	%
Turbine speed	6000	RPM
Synchronous generator speed	450	RPM
System design lifetime	25	yr

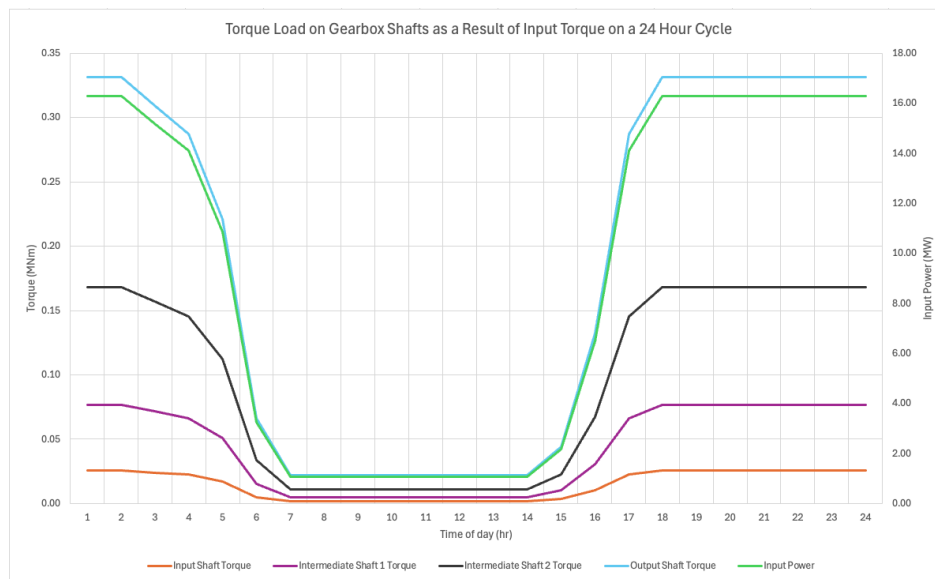


Figure 7. Torque Load on Gearbox Shafts as a Result of Input Torque on a 24 Hour Cycle

Table 3. *Varying Gear Speeds due to Electrical Generation Profile and Output Generation due to Efficiencies*

Time (24Hr)	Daily Electrical Generation (MW)	Actual Turbine Power Output (MW)	Shaft	Speed [rpm]	Speed [rad/s]	Torque with Efficiencies [MNm]
1-2	15	16.28	Input	6000.00	628.32	0.025904
			Shaft 2	2000.00	209.44	0.076662
			Shaft 3	900.00	94.25	0.168058
			Output	450.00	47.12	0.331573
3	14	15.19	Input	6000.00	628.32	0.024177
			Shaft 2	2000.00	209.44	0.071551
			Shaft 3	900.00	94.25	0.156854
			Output	450.00	47.12	0.309468
4	13	14.11	Input	6000.00	628.32	0.022450
			Shaft 2	2000.00	209.44	0.066440
			Shaft 3	900.00	94.25	0.145650
			Output	450.00	47.12	0.287363
5	10	10.85	Input	6000.00	628.32	0.017269
			Shaft 2	2000.00	209.44	0.051108
			Shaft 3	900.00	94.25	0.112038
			Output	450.00	47.12	0.221049
6	3	3.26	Input	6000.00	628.32	0.005181
			Shaft 2	2000.00	209.44	0.015332
			Shaft 3	900.00	94.25	0.033612
			Output	450.00	47.12	0.066315
7-14	1	1.09	Input	6000.00	628.32	0.001727
			Shaft 2	2000.00	209.44	0.005111
			Shaft 3	900.00	94.25	0.011204
			Output	450.00	47.12	0.022105
15	2	2.17	Input	6000.00	628.32	0.003454
			Shaft 2	2000.00	209.44	0.010222
			Shaft 3	900.00	94.25	0.022408
			Output	450.00	47.12	0.044210
16	6	6.51	Input	6000.00	628.32	0.010362
			Shaft 2	2000.00	209.44	0.030665
			Shaft 3	900.00	94.25	0.067223
			Output	450.00	47.12	0.132629
17	13	14.11	Input	6000.00	628.32	0.022450
			Shaft 2	2000.00	209.44	0.066440
			Shaft 3	900.00	94.25	0.145650
			Output	450.00	47.12	0.287363
18-24	15	16.28	Input	6000.00	628.32	0.025904
			Shaft 2	2000.00	209.44	0.076662
			Shaft 3	900.00	94.25	0.168058
			Output	450.00	47.12	0.331573

8.0 Gearbox Element Design

8.1 Gearbox Layouts, Drawings & Cross-sections

The following figures depict the layout of the gearbox elements, drawn to scale using SolidWorks. The bearings supporting each shaft can be seen, along with locking nuts on locating bearings and seals on the input and output shafts. The housing (not shown) will support each bearing from below, as well as clamp each one from above. The shaft seals will fit within a groove in the housing, with a fit such that the shafts are free to spin within the seal, and the seal remains stationary.

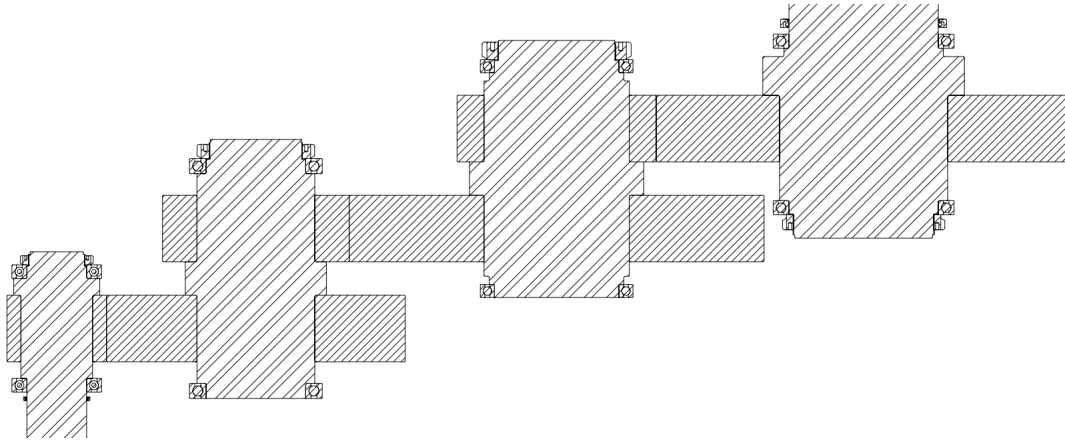


Figure 8. *Gear-box Cross-Section*

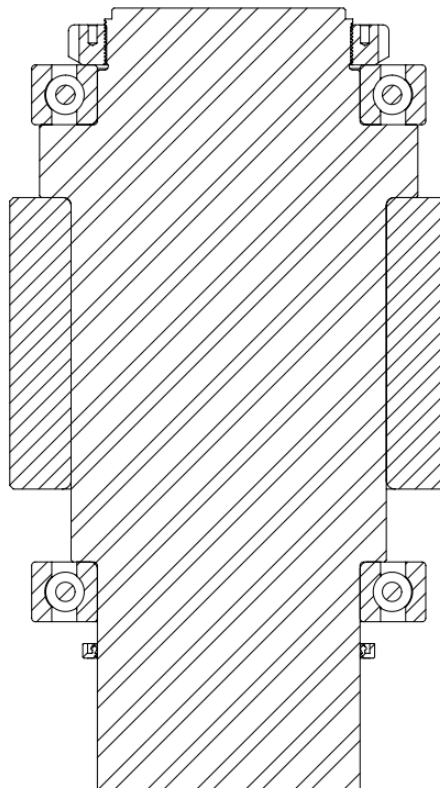


Figure 9. *Shaft 1 Cross Section*

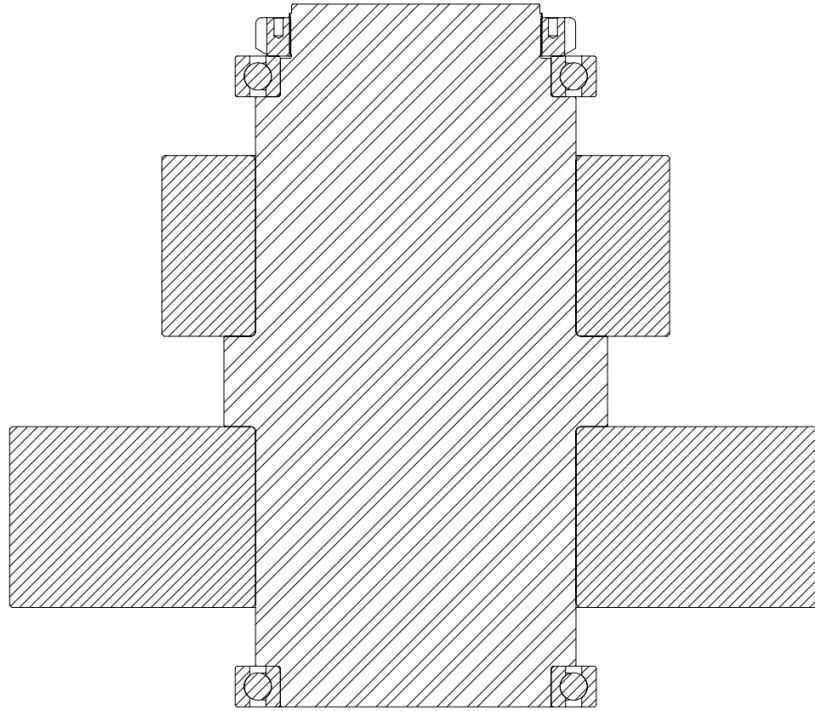


Figure 10. *Shaft 2 Cross Section*

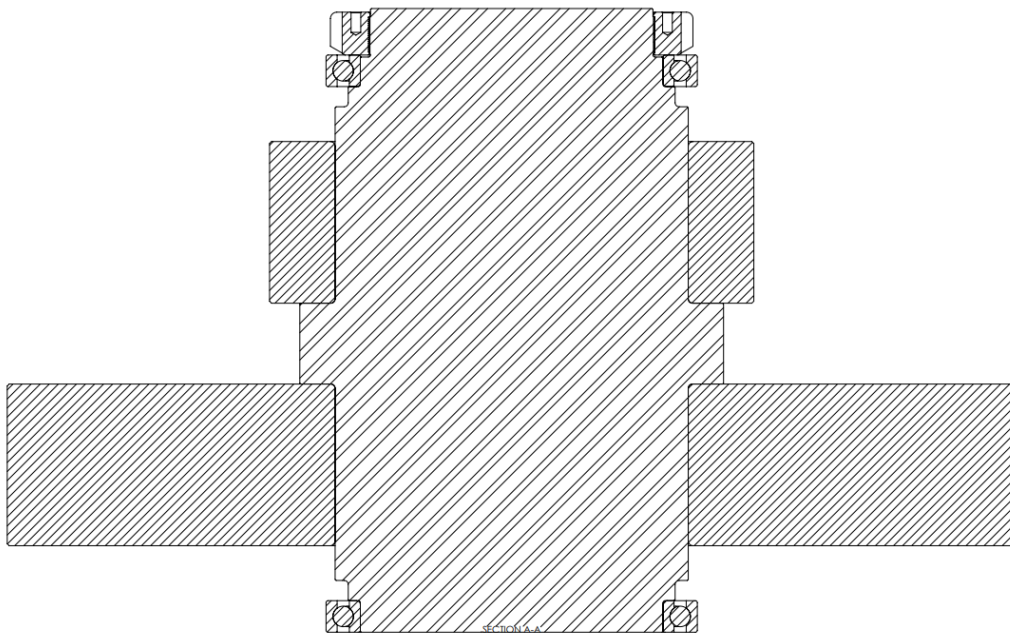


Figure 11. *Shaft 3 Cross Section*

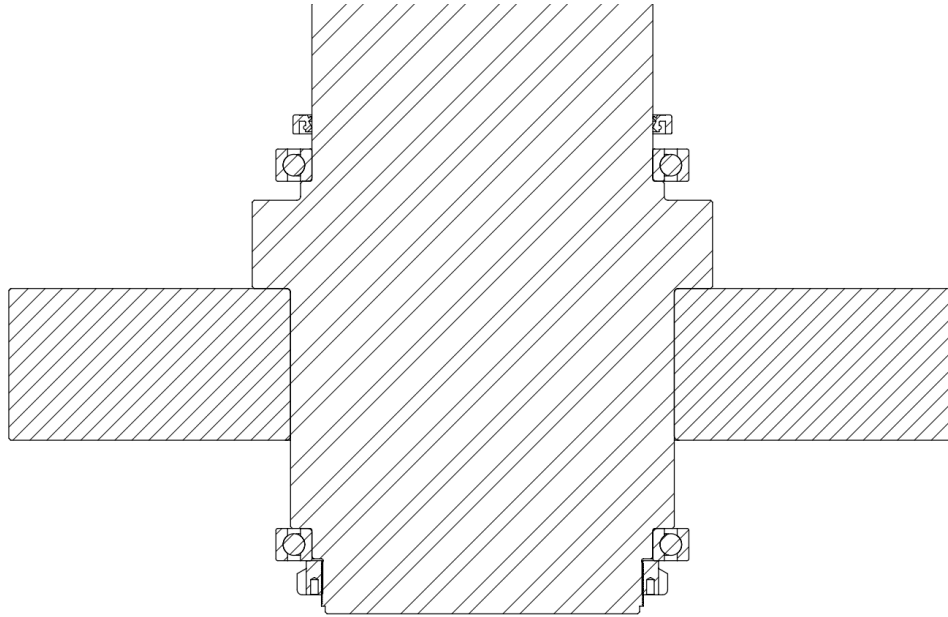


Figure 12. *Shaft 4 Cross Section*

8.2 Preliminary Sizing & Specifications

For the following sections, Figure 13 shows the naming and locations used throughout. Gears are labeled A through F starting at the input and ending at F at the output. The bearings are labeled A through H also starting at the input and ending at H at the output.

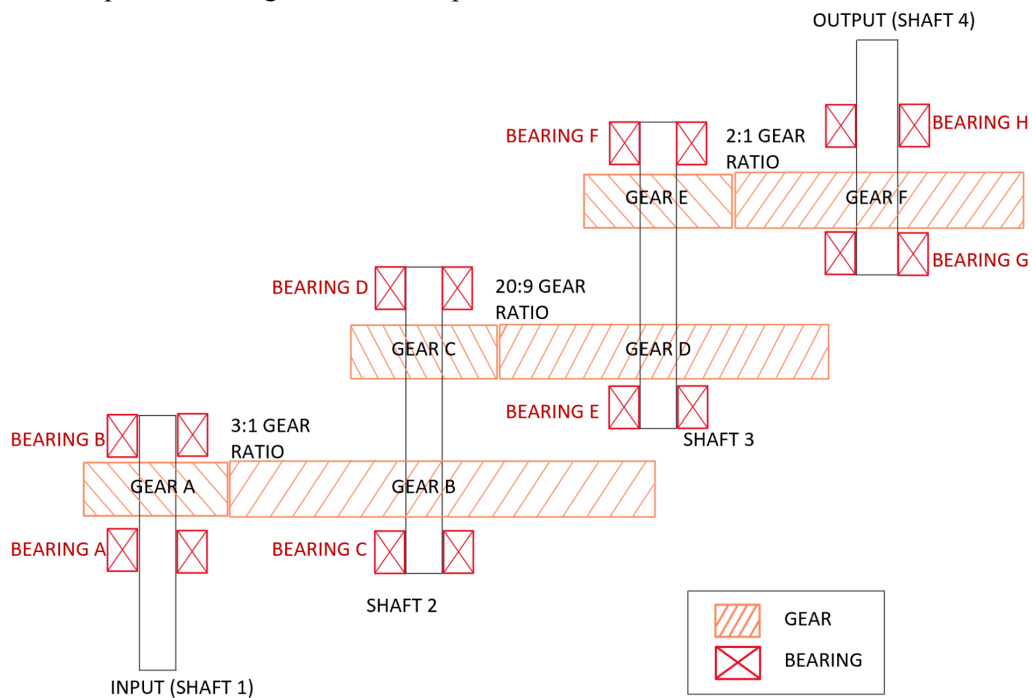


Figure 13. *Gear and Bearing Naming and Locations*

8.2.1 Gears

The specification of gear size is the first step in the design of individual components. As previously outlined in this report, the gearbox is required to receive and transfer a maximum of 16.28 MW, with an estimated efficiency of 96% across the unit. To specify material properties, the gear material was first selected to be **AISI 4140 nitrided steel**.

The torque acting on each shaft is found in Section 7, along with the speed at which each shaft will rotate. Using these set values, the number of teeth, face width, mesh module and pressure angle of each gear is selected. Note that the pressure angle, mesh module and face width must match for two meshed gears. Also note the number of teeth of the driven gear is determined by the preselected gear ratio for each meshing pair, and by selecting only the number of teeth of the pinion (driving gear). The selection of gear geometry was iterated on until an acceptable factor of safety was obtained.

Table 4 summarizes all gear specifications. A breakdown of the calculation of some critical values is followed, and all gear calculation sheets can be found in Appendix B.

Table 4. Gear Specification Summary

Gear Specification Summary						
	Gear A	Gear B	Gear C	Gear D	Gear E	Gear F
N - Number of Teeth	24	72	45	100	25	50
F - Face Width [m]	0.4	0.4	0.4	0.4	0.4	0.4
m - Mesh Module [mm]	25	25	25	25	50	50
Φ - Pressure Angle [°]	25	25	25	25	25	25
d - Diameter [m]	0.6	1.8	1.125	2.5	1.25	2.5
W _t - Tangential Force [kN]	86.37	86.37	136.32	136.32	268.96	268.96
S _{fb} - Corrected Bending-fatigue Strength [MPa]	183.8	183.8	183.8	183.8	183.8	183.8
S _{fc} - Corrected Surface Fatigue Strength [MPa]	747.1	747.1	747.1	747.1	747.1	747.1
σ_b - Bending Stress [MPa]	48.6	42.0	70.1	66.3	73.4	69.2
σ_c - Surface Stress [MPa]	430.7	430.7	396.4	396.4	550.1	550.1
N _b - Bending Stress Safety Factor	3.78	4.37	2.62	2.77	2.51	2.66
N _c - Surface Stress Safety Factor	3.01	3.01	3.55	3.55	1.84	1.84

The diameter of each gear is calculated by multiplying the number of teeth (N) by the mesh module (m):

$$d = mN$$

A critical intermediate value that is calculated is the tangential force (W_T) acting at the edge of each gear. The tangential force is found through applying the following equations, which use the previously found torques (T) of each shaft, as well as the newly defined diameter.

$$W_T = \frac{2T_p}{d_p}$$

Note that the equation above shows that W_T is only dependent on the torque and diameter of the pinion (driving gear) of each meshed pair. At this point, all relevant bending stress, fatigue stress, bending

strength and fatigue strength correction factors are found based on the defined gear geometry. All correction factors are outlined in the corresponding calculation sheets shown in Appendix B, along with a justification of each choice. Several key assumptions that are used to find correction factors are:

1. Operating temperature of the gearbox is below 250°F, and therefore has no effect on stress
2. All gears have a 99% reliability
3. Each gear is machined to meet a quality value of 12
4. The driving turbine and driven generator both produce uniform shock on the gear box

Bending stress (σ_b) and surface stress (σ_c) are then calculated using the following equations:

$$\sigma_b = \frac{W_t}{FmJ} \frac{K_a K_m}{K_v} K_s K_B K_I \quad \sigma_c = C_P \sqrt{\frac{W_t}{FId} \frac{C_a C_m}{C_v} C_s C_f}$$

Bending fatigue strength (S_{fb}) and surface fatigue strength (S_{fc}) were calculated as follows:

$$S_{fb} = \frac{K_L}{K_T K_R} S_{fb'} \quad S_{fc} = \frac{C_L C_H}{C_T C_R} S_{fc'}$$

8.2.2 Bearings

In theory, we would need to select bearings that would last 25 years undergoing 6000 rpm's at 15 MW. However, because we do not run our gearbox at 15 MW for 24 hours, we have varying torque on the shafts. If the turbine was providing 15 MW for 24 hours, the total torque on the first shaft would be equal to 0.777 MNm for 24 hours. Considering the varying power, the total torque after 24 hours is 0.1589 MNm (Summing the torques from Table 3). Therefore, when doing our bearing calculations we want to consider this to prevent overdesign and excessive costs.

The lifespan for 25 years at 15 MW will not be considered for 24 hours, but rather,

$$\frac{0.1589 \text{ MNm}}{x} = \frac{0.777 \text{ MNm}}{24 \text{ Hours}} \\ x = 4.9067 \text{ Hours}$$

We can effectively say, for the purpose of determining our required lifespan, that the first shaft will run at 15 MW, and torque of 0.025904 MNm, for 4.9067 Hours per day for 25 years.

Now that we have taken into account the varying torque loads, we can use this method to find the total number of life cycles required before maintenance for each shaft.

$$\text{Cycles}_{\text{Shaft 1}} = (4.9067 \text{ hours} * 60 \text{ min})(365 \text{ days})(25 \text{ years})(6000 \text{ rpm}) = 1.612E10$$

$$\text{Cycles}_{\text{Shaft 2}} = (4.9067 \text{ hours} * 60 \text{ min})(365 \text{ days})(25 \text{ years})(2000 \text{ rpm}) = 5.373E09$$

$$\text{Cycles}_{\text{Shaft 3}} = (4.9067 \text{ hours} * 60 \text{ min})(365 \text{ days})(25 \text{ years})(900 \text{ rpm}) = 2.418E09$$

$$Cycles_{Shaft 4} = (4.9067 \text{ hours} * 60 \text{ min})(365 \text{ days})(25 \text{ years})(450 \text{ rpm}) = 1.209E09$$

Based on the required cycles for each shaft, we used a trial and error method to select proper bearings on each shaft from SKF [11]. The torque used in the calculations is the maximum torque on each shaft, since we already accounted for variation in time (using 4.9067 hr instead of 24 hr). See Appendix for how the calculations were completed in excel for the shafts. Figure 14 shows the final selected bearings.

	a & b	c	d	e	f	g & h
Life Cycles Before Maintenance:	1.61E+10	5.37E+09	5.37E+09	2.42E+09	2.42E+09	1.21E+09
Calculated Basic Life Rating:	5.11E+10	6.65E+09	5.32E+10	1.30E+10	1.04E+11	4.26E+09
Good or Bad?	Good	Good	Good	Good	Good	Good
Designation on SKF.com	6072 M	619/600 MA	619/600 MA	618/750 MA	618/750 MA	618/900 MA

Figure 14. Bearing Selection from Calculations

8.2.3 Couplings

To select the appropriate couplings for the gearbox that connects the turbine to the input shaft and the output shaft to the generator, we must ensure they can handle the maximum torque loads.

The selection of couplings was based on the required information and methodology as recommended by SKF [12]. The process follows a structured approach to ensure that all operational requirements are met. The steps from SKF are as follows:

1. Determine system torque:
 - o The system torque was calculated by considering the maximum power transmitted and the operating speeds of the shafts,

$$T_{input} = \frac{P_{max}}{\omega_{input}} = \frac{(16,280 \text{ kW})(9550)}{6000 \text{ rpm}} = 25,912.3 \text{ Nm}$$

$$T_{output} = \frac{P_{max}}{\omega_{output}} = \frac{(16,280 \text{ kW})(9550)}{450 \text{ rpm}} = 345,497.8 \text{ Nm}$$

2. Service factor:
 - o According to Table 9 on page 87 from SKF [12], the service factor for turbines is 1.25 and the service factor for generators is 1.00.

3. Required coupling rating:
 - o The required coupling rating was calculated by multiplying the system torque by the service factor,

$$Required \text{ Coupling Rating}(input) = T_{input} \times Service \text{ Factor} = 32,390.4 \text{ Nm}$$

$$Required \text{ Coupling Rating}(output) = T_{output} \times Service \text{ Factor} = 345,497.8 \text{ Nm}$$

4. Coupling selection:
 - o Using SKF's product selection guide, the appropriate couplings were chosen based on the calculated required rating, but also selected to accommodate the physical dimensions of the shafts.

The selected input coupling is **1150 TGH** (rated for 39,800 Nm of torque) and the output coupling is **1230 TGH** (rated for 435,000 Nm of torque).

8.2.4 Lock Nut and Clip

In order to lock the “fixed” bearing on each shaft into its position, we need to secure a lock nut and locking clip onto the shaft against the bearing. The “floating” bearing will not be secured with a lock nut, but just the housing. For each of the floating bearings on each of the shafts, we have selected a lock nut and locking clip for the dimensions of our shafts from SKF [14]. See Table 5 below.

Table 5. Lock nut selection for fixed bearings

Fixed Bearing on Each Shaft	Selected Lock Nut	Locking Clip
Bearing B	HM 3168	MS 3168
Bearing D	HM 31/560	MS 31/600-560
Bearing F	HM 31/710	MS 31/710
Bearing G	HM 30/850	MS 30/900-850

8.2.5 Seals

There are two major spots in the gearbox that require industrial power transmission seals, which include the contact points between the input shaft and the housing, and the output shaft and housing. From SKF [15], we selected Radial Shaft Seals as these are designed to sit between a rotating and non-rotating component. The selected seals are shown in Table 6 below.

Table 6. Seal selection

Location	Selected Seal
Input Shaft and Housing	360X400X20 HMSA10 V
Output Shaft and Housing	900X1000X20 HMSA10 V

8.2.6 Housing

As previously described, the housing will consist of a top and bottom shell. The bottom shell will be fixed to the mounting pad by bolting a base flange down, and the top shell will be bolted to the bottom shell. The length of the interior of the gearbox is required to be at minimum 6.5m, which is the distance between the outermost extremities of the gear A to gear F. Some additional clearance will be added to ensure the gears never contact the housing. The normal distance between each seal is 2.3m, thus this is the minimum required width of the gearbox. Note that the final width may be up to one meter larger than this distance, to allow for a suitable wall thickness and clearance on all parts.

All shafts will be held in the same horizontal plane as one another. As the largest gear is 2.5m in diameter, the plane must be at minimum 1.25m from the ground, plus room for ground clearance. To maintain a nominal number for the shaft centerline, this plane will be at approximately 1.75m from the base, allowing for 0.5m of clearance. This is measured from the ground, and thus to find the actual clearance of the largest gear, the base thickness of the housing would be subtracted from the clearance value (0.5m).

8.3 Potentially Critical Points

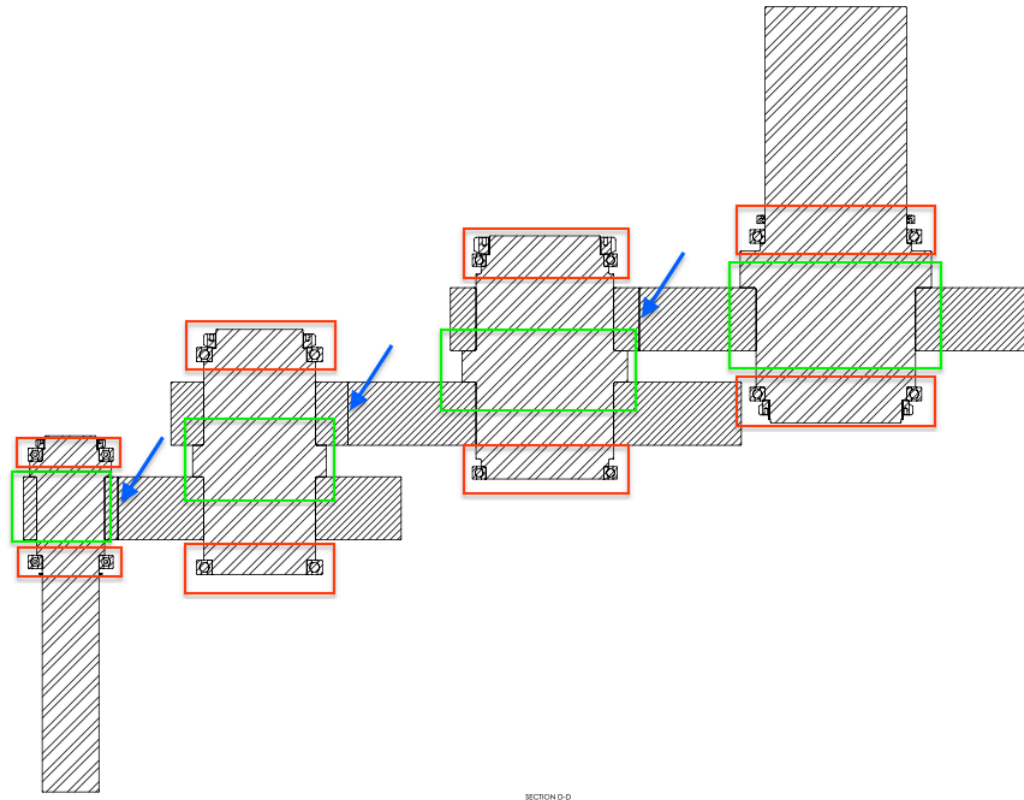


Figure 15. *Potential Critical Locations*

Bearings installed at various points (red box) along the shafts are crucial as they handle the radial loads which induce significant stress. These bearings are essential for maintaining the stability and alignment of the shafts. Failures in this component can lead to severe issues like excessive shaft play, impacting the gearbox's functionality. Regular monitoring for signs of wear and proper lubrication are key strategies to extend the lifespan and efficiency of the bearings.

The tapered sections of the shafts (green box), where gears are mounted, are susceptible to high stress concentrations due to the transition in diameter. This design feature, while beneficial for ensuring a tight fit of the gears, can lead to increased stress under load, especially where significant torque is transmitted.

The gear meshes (blue arrow) are subject to high contact stresses, making them vulnerable to wear and damage such as pitting, spalling, or tooth breakage. Ensuring these points are well-aligned, lubricated, and

made from suitable materials is essential. Any imperfections in tooth form or alignment at these critical points can significantly impact the gearbox's performance.

9.0 Preliminary Design for Shafts

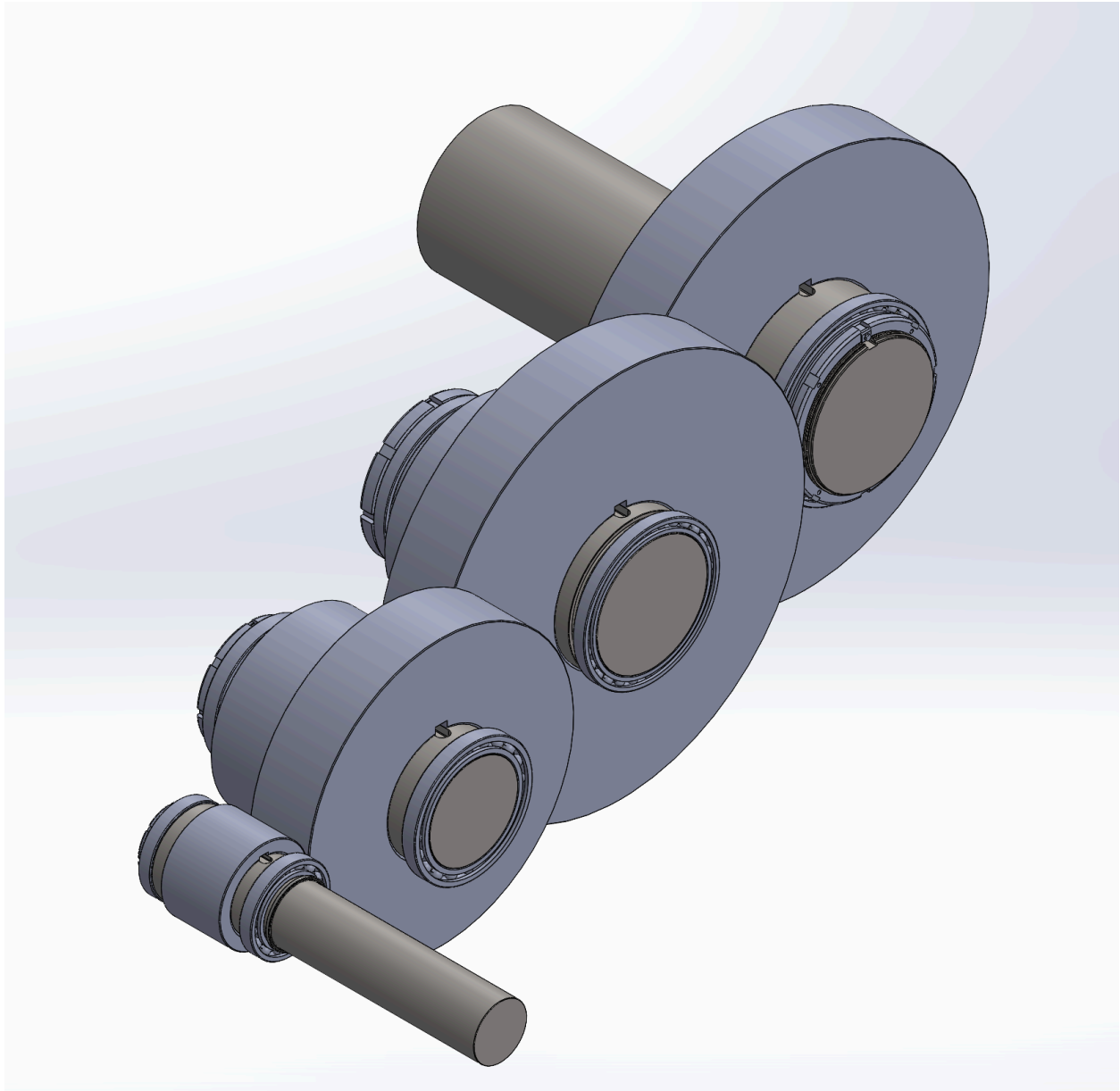


Figure 16: Gear Box Assembly, cylinders represent pitch diameter and face width

Each shaft in our design shared a few key features. Each was designed to have two bearings, a key for each gear connection, and varying sizes of steps to allow for required assembly processes. All features can be seen in Figure 16 above. In order to secure each of the locating bearings a locking nut was added. This required yet another step that will be threaded to accommodate the locking nut thread. Each keyway was chosen based on in class examples, scaled to a relevant size.

To Design a shaft that will last for the expected lifetime of the project there are multiple factors that go into its design. When our team was designing the shafts we started by find the corrected endurance limit S_e of 4140 Steel ($S_{ut} = 1772 \text{ MPa}$). Since the ultimate strength is greater than 1400 MPa, the uncorrected endurance limit of our material is $S_e' = 700 \text{ MPa}$. Using the correction factors determined in Appendix C, we calculated the corrected endurance limit of our material using the following equation.

$$S_e = C_{load} C_{size} C_{surf} C_{temp} C_{reliab} S_e'$$

Once found, we created FBD's and bending moment diagrams (See Appendix C). Using those diagrams as well as our preliminary notch shaft designs, we determined the critical points of the shaft as shown in sections 9.1 - 9.4. After selecting our critical points, we calculate the max bending moment stresses and torsion for each shafts critical points in terms of diameter using the following equations:

$$\sigma_{max,nom} = \frac{Md}{2I}$$

$$\tau_{max,nom} = \frac{Td}{2J}$$

We then chose arbitrary diameter ratios that we used to determine the stress concentration factor, notch sensitivity, fatigue stress concentration factor. Using Neuber's constant ($\sqrt{a}$) of our materials ultimate strength, we determined our notch sensitivity. We did this for both shear and bending cases.

$$K_t = A \left(\frac{r}{d} \right)^b$$

$$q = \frac{1}{1 + \frac{\sqrt{a}}{\sqrt{r}}}$$

$$K_f = 1 + q(K_t - 1)$$

We then modified our nominal stress using the fatigue stress concentration factor.

$$\sigma_{max} = K_f \sigma_{nom}$$

$$\tau_{max} = K_{fs} \tau_{nom}$$

Using the determined stress values at the critical points we did fatigue analysis to determine the appropriate Von Mises stress (σ').

$$\sigma' = \sqrt{\sigma_{max}^2 + 3\tau_{max}^2}$$

Finally, we assumed that the shaft would have infinite life so we divided the corrected safety factor by the Von Mises stress to obtain a Safety factor in between 1.5-2.5.

$$SF = \frac{S_e}{\sigma'}$$

If the SF was not in those bounds we would iteratively try different diameters until we found one that worked.

9.1 Shaft FBD and Calculations

For each shaft we started by calculating the reaction forces on each bearing and force of the gears. Once the forces were found we then moved on to finding the torques transferred by the gears on to each shaft. Once all the torques were found we performed shear and bending moment diagrams. The following figures below show the free body diagrams of each shaft. The shear and bending moment diagrams can be found in appendix C.

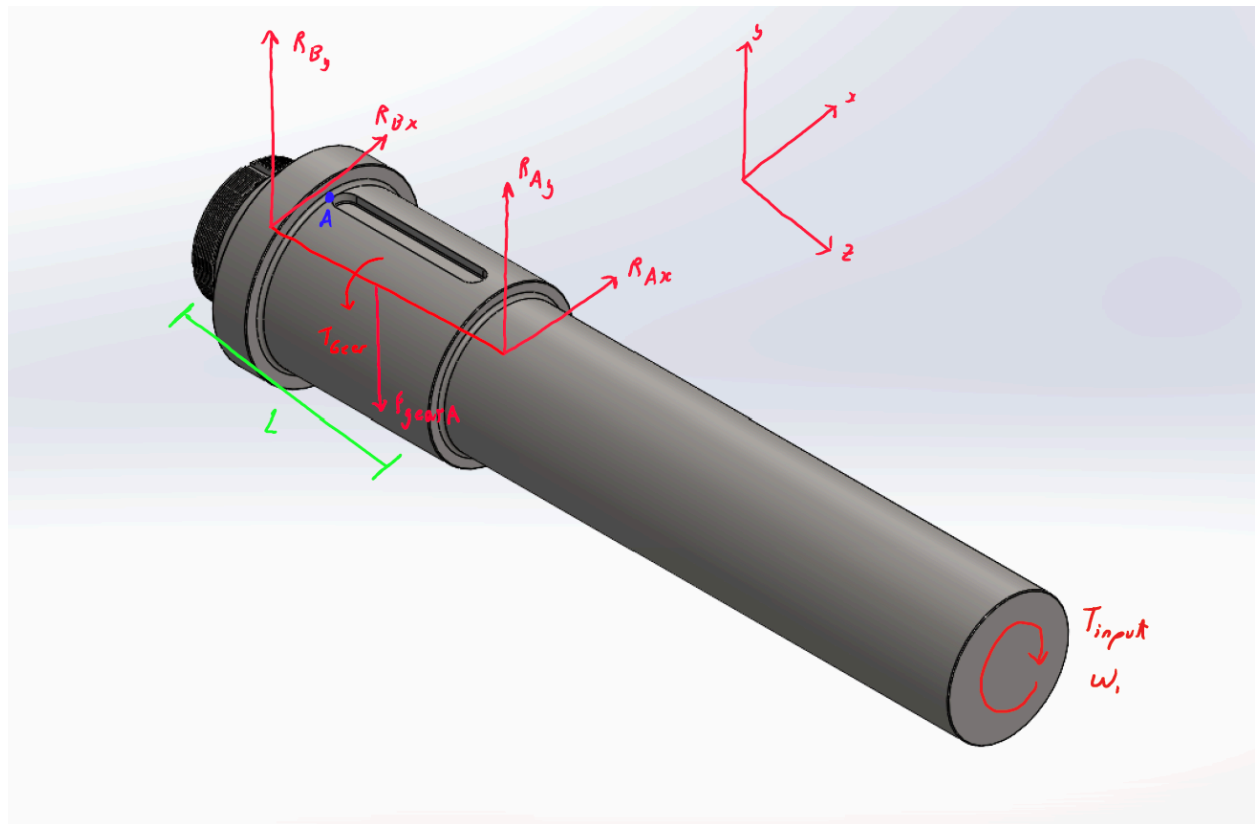


Figure 17: Shaft 1 FBD

Static Loadings Calculations for shaft 1:

$$\begin{aligned}\Sigma F_x &= 0 = R_{bx} + R_{ax} \\ \Sigma F_y &= 0 = F_{gearA} + R_{by} + R_{ay} \\ \Sigma M_{xa} &= 0 = LR_{by} + \frac{L}{2}F_{gearA} \\ R_{by} &= -\frac{1}{2}F_{gearA} \\ R_{ay} &= -\frac{1}{2}F_{gearA}\end{aligned}$$

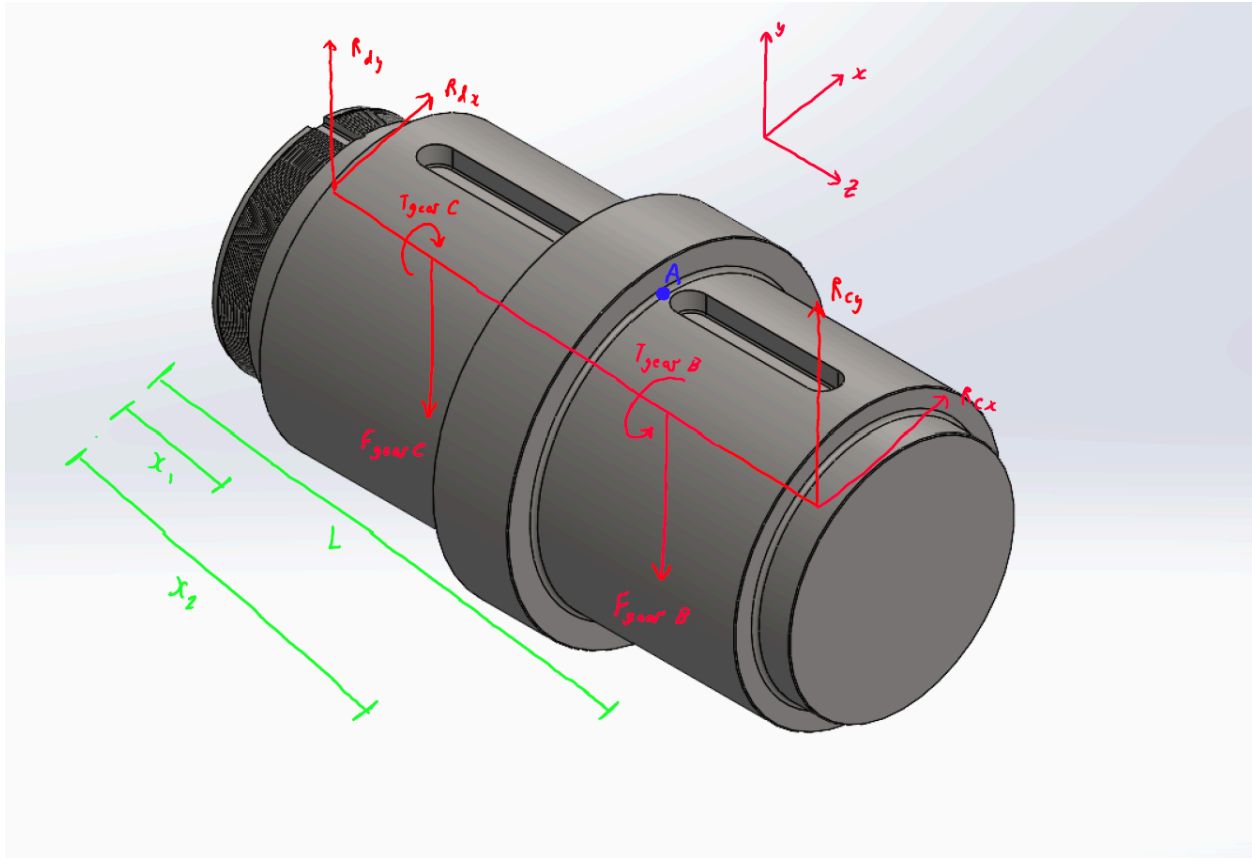


Figure 18: Shaft 2 FBD

Static Loading Calculations for shaft 2:

$$\begin{aligned}\Sigma F_y &= 0 = F_{gearC} - F_{gearB} + R_{dy} + R_{cy} \\ \Sigma M_{xd} &= 0 = -x_1 F_{gearC} - x_2 F_{gearB} + LR_{cy} \\ R_{cy} &= F_{gearC} \left(\frac{x_1}{L} \right) + F_{gearB} \left(\frac{x_2}{L} \right) \\ R_{dy} &= F_{gearC} + F_{gearB} - R_{cy}\end{aligned}$$

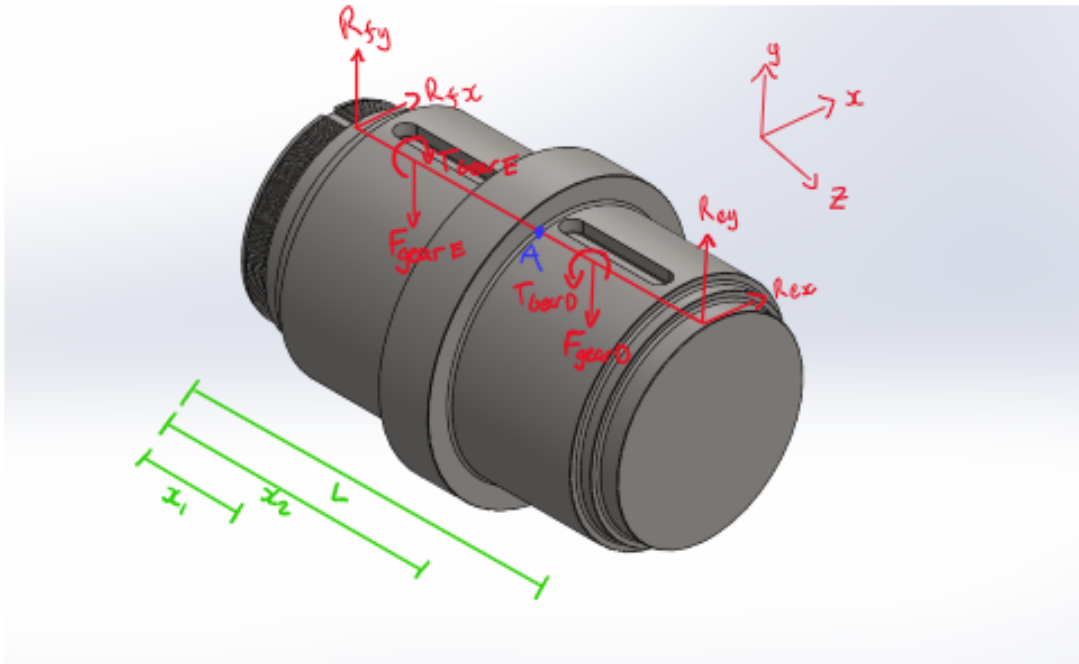


Figure 19: Shaft 3 FBD

Static Loading Calculations for shaft 3:

$$\begin{aligned}\Sigma F_y = 0 &= R_{ey} - F_{gearD} - F_{gearE} + R_{fy} \\ \Sigma M_{xh} = 0 &= -x_1 F_{gearE} - x_2 F_{gearD} + L R_{ey} \\ R_{ey} &= F_{gearE} \left(\frac{x_1}{L} \right) + F_{gearD} \left(\frac{x_2}{L} \right) \\ R_{fy} &= F_{gearD} + F_{gearE} - R_{ey}\end{aligned}$$

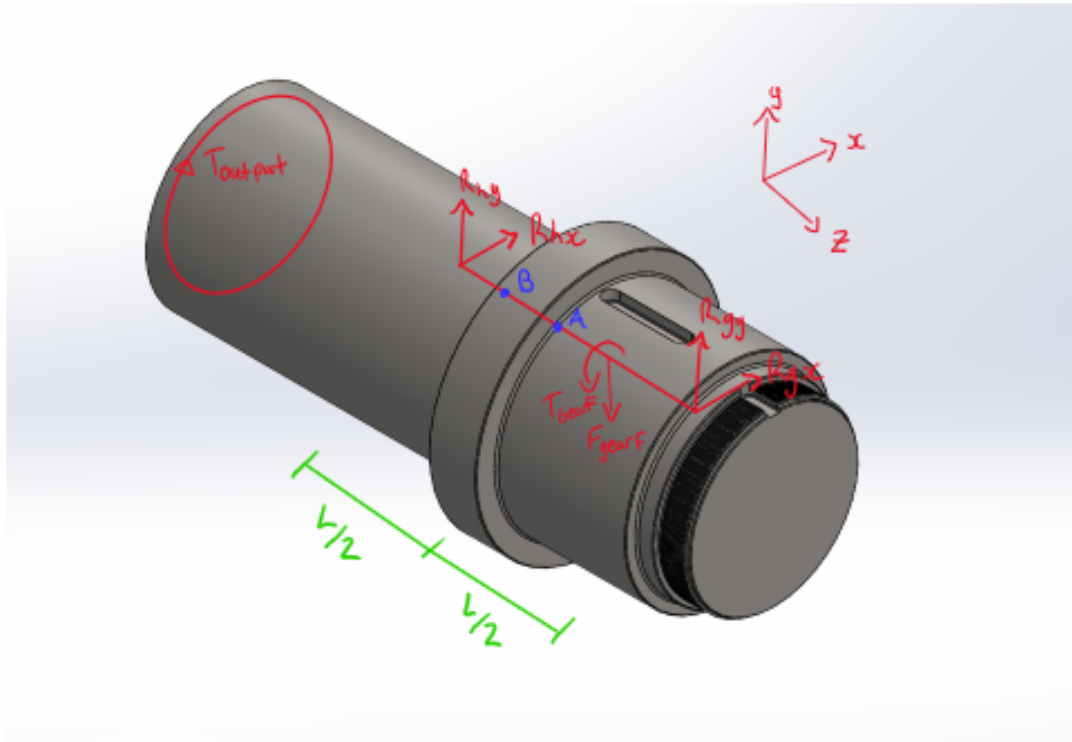


Figure 20: Shaft 4 FBD

Static Loading Calculations for shaft 4:

$$\Sigma F_y = 0 = R_{hx} + R_{gx} - F_{gearF}$$

$$\Sigma M_{xh} = 0 = LR_{gy} - \frac{L}{2}F_{gearF}$$

$$R_{gy} = \frac{1}{2}F_{gearF}$$

$$R_{hy} = \frac{1}{2}F_{gearF}$$

10.0 Assembly Process

The broader approach to assembling the gearbox is demonstrated in Table 7 below.

Table 7. Process to Assemble Gearbox [13]

Steps	Component	Details
1	Housing Preparation	Clean and inspect housing for any defects. Apply lubricant to inner surfaces of housing.
2	Shaft Preparation	Install bearings on the shafts. Install gears and spacers on the shafts as per the specifications
3	Shaft Installation	Place the assembled shafts into the gearbox housing. Ensure correct alignment and seating.
4	Bearing Adjustment	Adjust bearing preload. Ensure the bearings are seated properly and preload is set according to the specification.
5	Seal Installation	Install oil seals at the shaft entries and exits. Lubricate seals and place gaskets on covers.
6	Final Assembly	Close the housing with covers, ensuring all seals and gaskets are in place. Apply sealant where necessary and tighten all bolts to specified torque.

11.0 Concluding Remarks and Next Steps

In this report, the conceptual design of a 15 MW gearbox for a concentrating solar-thermal power (CSP) system is expanded upon. A 3-stage gearbox design is selected to accommodate the required 40:3 overall ratio. Using the chosen gear ratios, operational shaft speeds, torque loads and power values were calculated for each shaft. A dynamic model of the shaft loading was developed based on the provided power output scheme to reflect the actual use-case.

Preliminary values for the number of teeth, face width, mesh module and pressure angle are selected for each gear. Using standard stress correction factors and fatigue strength correction factors based on the gear loading, the bending stress safety factor and surface stress safety factor are calculated. The gear specification process is iterated on until adequate safety factors are achieved, with the lowest being 1.84. The selected arrangement has the smallest gear as 0.6m in diameter, and the largest as 2.5m in diameter.

Shaft design follows the specification of gears. The input shafts each support one gear, while both intermediate shafts support two gears. All shafts have a series of steps to create shoulders for gears and bearings. Bearing lock nuts and washers have been specified and are used to lock the axial position of the bearing along the shaft. The shaft diameters vary between 0.36m and 1.35m. The lowest safety factor of all shafts is 1.52. Each shaft is supported by a ball bearing on each end, which are specified to ensure they support the load for the lifetime of the gearbox (25 years). Additionally, couplers are specified to connect the gearbox input and output shafts to their respective external components.

Moving forward, our team will seek input from Dr. Nikolai Dechev and his team to validate the gearbox design and calculations outlined in this report. Upon validation and feedback, our team will continue to refine the existing design, and develop detailed engineering drawings for manufacturing.

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Appendix A. Bearing Data Sheets and Calculations

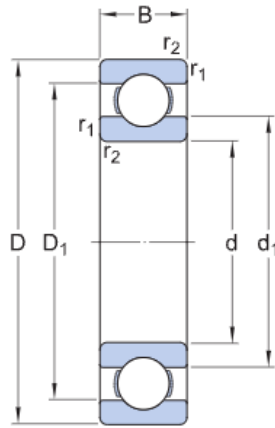


Figure A-1. Bearing Dimensions

Bearing A & B Datasheet:

Bearing 6072 M	
Properties	
Bore type	Cylindrical
Material, bearing	Bearing steel
Number of rows	1
Dimensions	
Bore diameter, d	360 mm
Outside diameter, D	540 mm
Width, B	82 mm
Shoulder diameter, d ₁	416.2 mm
Shoulder diameter, D ₁	484.78 mm
Chamfer dimension, r _{1,2}	min. 5 mm
Calculation data	
Basic dynamic load rating, C	442 kN
Basic static load rating, C ₀	695 kN

Fatigue load limit, P_u	14 kN
Reference speed	2400 rpm
Limiting speed	1900 rpm
Minimum Load Factor, k_r	0.025
Calculation factor, f_o	15.9

Table A-1. Bearing A & B Datasheet

Bearing C & D Datasheet:

Bearing 619/600 MA	
Properties	
Bore type	Cylindrical
Material, bearing	Bearing steel
Number of rows	1
Dimensions	
Bore diameter, d	600 mm
Outside diameter, D	800 mm
Width, B	90 mm
Shoulder diameter, d_1	663.8 mm
Shoulder diameter, D_1	735.67 mm
Chamfer dimension, $r_{1,2}$	min. 5 mm
Calculation data	
Basic dynamic load rating, C	585 kN
Basic static load rating, C_o	1220 kN
Fatigue load limit, P_u	19.6 kN
Reference speed	1400 rpm
Limiting speed	1100 rpm
Minimum Load Factor, k_r	0.02

Calculation factor, f_o	16.7
---------------------------	------

Table A-2. Bearing C & D Datasheet

Bearing E & F Datasheet:

Bearing 618/750 MA	
Properties	
Bore type	Cylindrical
Material, bearing	Bearing steel
Number of rows	1
Dimensions	
Bore diameter, d	750 mm
Outside diameter, D	920 mm
Width, B	78 mm
Shoulder diameter, d_1	804.5 mm
Shoulder diameter, D_1	865.02 mm
Chamfer dimension, $r_{1,2}$	min. 5 mm
Calculation data	
Basic dynamic load rating, C	527 kN
Basic static load rating, C_o	1250 kN
Fatigue load limit, P_u	18.3
Reference speed	1100 rpm
Limiting speed	900 rpm
Minimum Load Factor, k_r	0.015
Calculation factor, f_o	17.4

Table A-3. Bearing E & F Datasheet

Bearing G & H Datasheet:

Bearing 618/900 MA	
Properties	
Bore type	Cylindrical
Material, bearing	Bearing steel
Number of rows	1
Dimensions	
Bore diameter, d	900 mm
Outside diameter, D	1090 mm
Width, B	85 mm
Shoulder diameter, d ₁	960.7 mm
Shoulder diameter, D ₁	1028.6 mm
Chamfer dimension, r _{1,2}	min. 5 mm
Calculation data	
Basic dynamic load rating, C	618 kN
Basic static load rating, C ₀	1600 kN
Fatigue load limit, P _u	21.6 kN
Reference speed	850 rpm
Limiting speed	700 rpm
Minimum Load Factor, k _r	0.015
Calculation factor, f ₀	17.5

Table A-4. Bearing G & H Datasheet

Bearing A & B Excel Calculations:

SHAFT 1:		
Bearing a & b:		
Step 1) Choose a Bearing from SKF.com		
Designation on SKF.com	Enter Value: 6072 M	Units:
Basic Static Load Rating C_0 :	695000	N
Calculation factor f_d :	15.9	
Basic dynamic load rating C:	442000	N
Bore Diameter, B	0.3600	m
Step 2) Determine limiting value e		
Diameter of Shaft 1, D:	0.36	m
Is $D \leq B$:	YES	
Torque on Shaft 1, T:	25904	N-m
Radius of Gear A, r:	0.6	m
Tangential Force W_t :	43173.33333	N
Pressure Angle:	25	degrees
Radial Force W_r :	20132.05596	N
Helix Angle:	0	degrees
Axial Force W_a :	0	N
Axial load, F_a , on one bearing:	0	N
Ratio for Corresponding e:	0.0000	
Corresponding e from Table 1:	0.0000	
Step 3) Determine Equivalent Dynamic Bearing Load, P		
V Factor:	1	
Length of Shaft:	0.67	m
Reaction Force on Bearing A in z:	-21586.66667	N
Reaction Force on Bearing A in y:	10066.028	N
Radial Load, F_r :	11909.126	N
Ratio of Axial and Radial Loads, R:	0	
If $R \leq e$, P:	11909.126	
If $R > e$, P:		
Radial Load Factor, X (Table 1):	1.000	
Axial Load Factor, Y (Table 1):	0.000	
Dynamic Bearing Load P:	11909.126	N
Step 4) Calculate Life Rating		
Exponent for Life Equation, p:	3	
Basic Life Rating, L_{10} :	5.11E+10	Revolutions

Figure A-2. Bearing A & B Excel Calculations

Bearing C & D and Bearing E & F Excel Calculations:

SHAFT 2:				SHAFT 3:			
Bearing c & d:				Bearing e & f:			
Step 1) Choose a Bearing from SKF.com				Step 1) Choose a Bearing from SKF.com			
Designation on SKF.com	Enter Value: 619/600 MA	Units:	Notes:	Designation on SKF.com	Enter Value: 618/750 MA	Units:	
Basic Static Load Rating C_0 :	1220000	N		Basic Static Load Rating C_0 :	1250000	N	
Calculation factor f_0 :	16.7			Calculation factor f_0 :	17.4		
Basic dynamic load rating C:	585000	N		Basic dynamic load rating C:	527000	N	
Bore Diameter, B	0.6000	m		Bore Diameter, B	0.7500	m	
Step 2) Determine limiting value e				Step 2) Determine limiting value e			
Diameter of Shaft 2:	0.6	m		Diameter of Shaft 3:	0.75	m	
Is $D \leq B$:	NO			Is $D \leq B$:	YES		
Torque on Shaft 2, T:	152285	N-m		Torque on Shaft 2, T:	152285	N-m	
Radius of Gear B, r:	1.8	m		Radius of Gear D, r:	2.5	m	
Tangential Force W_t :	84602.77778	N		Tangential Force W_t :	60914	N	
Pressure Angle:	25	degrees		Pressure Angle:	25	degrees	
Radial Force W_r :	39450.92318	N		Radial Force W_r :	28404.66469	N	
Helix Angle:	0	degrees		Helix Angle:	0	degrees	
Axial Force W_a :	0	N		Axial Force W_a :	0	N	
Axial load, F_a , on one bearing:	0	N		Axial load, F_a , on one bearing:	0	N	
Ratio for Corresponding e:	0.0000			Ratio for Corresponding e:	0.0000		
Corresponding e from Table 1:	0			Corresponding e from Table 1:	0		
Step 3) Determine Equivalent Dynamic Bearing Load, P				Step 3) Determine Equivalent Dynamic Bearing Load, P			
V Factor:	1			V Factor:	1		
Length of Shaft:	1.35	m		Length of Shaft:	1.35	m	
Bearing c:				Bearing e:			
Reaction Force on Bearing c in z:	56401.85185	N		Reaction Force on Bearing e in z:	40609.33333	N	
Reaction Force on Bearing c in y:	26300.615	N		Reaction Force on Bearing e in y:	18936.443	N	
Radial Load, F_r :	31116.279	N		Radial Load, F_r :	22403.721	N	
Ratio of Axial and Radial Loads:	0			Ratio of Axial and Radial Loads:	0		
If $R < e$, P:	31116.279			If $R < e$, P:	22403.721		
If $R > e$, P:				If $R > e$, P:			
Radial Load Factor, X (Table 1):	1.000			Radial Load Factor, X (Table 1):	1.000		
Axial Load Factor, Y (Table 1):	0.000			Axial Load Factor, Y (Table 1):	0.000		
Dynamic Bearing Load P:	31116.279	N		Dynamic Bearing Load P:	22403.721	N	
Bearing d:				Bearing f:			
Reaction Force on Bearing d in z:	28200.92593	N		Reaction Force on Bearing f in z:	20304.66667	N	
Reaction Force on Bearing d in y:	13150.308	N		Reaction Force on Bearing f in y:	9468.222	N	
Radial Load, F_r :	15558.139	N		Radial Load, F_r :	11201.860	N	
Ratio of Axial and Radial Loads:	0			Ratio of Axial and Radial Loads:	0		
If $R < e$, P:	15558.139			If $R < e$, P:	11201.860		
If $R > e$, P:				If $R > e$, P:			
Radial Load Factor, X (Table 1):	1.000			Radial Load Factor, X (Table 1):	1.000		
Axial Load Factor, Y (Table 1):	0.000			Axial Load Factor, Y (Table 1):	0.000		
Dynamic Bearing Load P:	15558.139	N		Dynamic Bearing Load P:	11201.860	N	
Step 4) Calculate Life Rating				Step 4) Calculate Life Rating			
Exponent for Life Equation, p:	3			Exponent for Life Equation, p:	3		
Basic Life Rating, L_{10} , Bearing c:	6.65E+09	Revolutions		Basic Life Rating, L_{10} , Bearing e:	1.30E+10	Revolutions	
Basic Life Rating, L_{10} , Bearing d:	5.32E+10	Revolutions		Basic Life Rating, L_{10} , Bearing f:	1.04E+11	Revolutions	

Figure A-3. Bearing C & D and Bearing E & F Excel Calculations

Bearing G & H Excel Calculations:

SHAFT 4:		
Bearing g & h:		
Step 1) Choose a Bearing from SKF.com		
	Enter Value:	Units:
Designation on SKF.com	618/900 MA	
Basic Static Load Rating C_0 :	1600000	N
Calculation factor f_0 :	17.5	
Basic dynamic load rating C:	618000	N
Bore Diameter, B	0.9000	m
Step 2) Determine limiting value e		
Diameter of Shaft 4:	0.9	m
Is $D \leq B$:	YES	
Torque on Shaft 3, T:	345388	N-m
Radius of Gear F, r:	2.5	m
Tangential Force W_t :	138155.2	N
Pressure Angle:	25	degrees
Radial Force W_r :	64422.82777	N
Helix Angle:	0	degrees
Axial Force W_a :	0	N
Axial load, F_a , on one bearing:	0	N
Ratio for Corresponding e:	0	
Corresponding e from Table 1:	0	
Step 3) Determine Equivalent Dynamic Bearing Load, P		
V Factor:	1	
Length of Shaft:	1	m
Reaction Force on Bearing g in z:	-59077.6	N
Reaction Force on Bearing g in y:	32211.414	N
Radial Load, F_r :	38109.349	N
Ratio of Axial and Radial Loads:	0	
If $R < e$, P:	38109.349	
If $R > e$, P:		
Radial Load Factor, X (Table 1):	1.000	
Axial Load Factor, Y (Table 1):	0.000	
Dynamic Bearing Load P:	38109.349	N
Step 4) Calculate Life Rating		
Exponent for Life Equation, p:	3	
Basic Life Rating, L_{10} :	4.26E+09	Revolutions

Figure A-4. Bearing G & H Excel Calculations

Appendix B. Gear Stress Calculation Sheets

Gear Stress Calculations - Gears A & B (1:3)				Legend:	Selected Value	Lookup Value	Calculated	Predetermined
DO NOT MODIFY								
Category	Variable	Gear A	Gear B	Notes & equations				
Gear Specs	N - Number of teeth	24	72	Gear Ratio: 1:3				
	F - Face Width (m)	0.4	0.4	Acceptable range (in):				
	m - Mesh Module (m)	0.025	0.025					
	Φ - Pressure Angle (°)	25	25					
	Tp - Pinion Torque (Nm)	25910	25910	Tp = P/ω				
	ω - Shaft Speed (rad/s)	628.32	209.44					
	d - Diameter (m)	0.6	1.8	d = N*m				
	R - Pitch Radius (in)	11.81	35.43	R = d/2				
	Pd - Diametral Pitch (in ⁻¹)	1.02	1.02	Pd ~ (1/m)				
	Wt - Tangential Force (N)	86366.67	86366.67	Wt = 2(Tp)/(Dp)				
	C - Center Distance (in)	47.24	47.24	C = R1 + R2				
	Xp - Pinion Addendum Coefficient	0.00	0.00	Assume = 0 (full-depth teeth)				
Bending Stress Correction	ρ - Radius of Curvature (in)	4.21	15.76	Eq. 12.22b				
	Vp - Pitch Velocity (m/s)	188.496	188.496	Vp = ω*(Dp/2)				
	J - Bending Geometry Factor	0.32	0.37	Table 12-12: Interpolated				
	Ka - Application Factor	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)				
	Km - Distribution Factor	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m				
	Kv - Dynamic Factor	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)				
	Qv - Quality Index	12	12	Designated as 12, based on Table 12-6				
	Ks - Size Factor	1	1	Assumed gear tooth size has no influence				
	Kb - Backup Ratio	1	1	All solid gears				
	Ki - Idler Factor	1	1	Both non-idling				
	Cp - Elastic Coefficient (MPa ^{1/2})	191	191	Table 12-18: Steel pinion, steel gear				
	I - Surface Geometry Factor	0.127	0.127	Eq. 12.22a				
Surface Stress Correction	Ca	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)				
	Cm	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m				
	Cv	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)				
	Cs	1	1	Assumed gear tooth size has no influence				
	Cf - Surface Finish Factor	1	1	Surface finish to be specified to meet requirement				
	Sfb' - Uncorrect. Bending-Fatigue Strength (Pa)	230000000	230000000	Selected Material: AISI 4140 Nitried				
	Kt - Temperature Factor	1	1	Assume T < 250F				
	Kr - Reliability	1	1	Assume 99% Reliability				
	Kl - Lifetime Factor	0.799	0.799	Figure 12-24, Low estimate for ~10 ⁷ load cycles				
	Sfb - Corrected Bending-Fatigue Strength (Pa)	183778858.4	183778858.4	Eq. 12.24a				
	Sfc' - Uncorrect. Surface Fatigue Strength (Pa)	1100000000	1100000000	AISI 4140 Nitried: 1100-1250MPa (table 12-21)				
	Cl - Lifetime Factor	0.679	0.679	Figure 12-26, Low estimate for ~10 ⁷ load cycles				
Surface Strength	Ch - Hardness Difference Factor	1	1	No difference in hardness				
	Ct - Temperature Factor	1	1	Assume T < 250F				
	Cr - Reliability Factor	1	1	Assume 99% Reliability				
	Sfc - Corrected Surface Fatigue (Pa)	747112078.1	747112078.1	Eq. 12.25				
Category	Output Values	Gear A	Gear B	Notes & equations				
Bending Stress	Bending Stress (Pa)	48581250	42016216.22	Eq. 12.15si				
Stress	Safety Factor	3.78	4.37					
Surface Stress	Surface Stress (Pa)	430670757.47	430670757.47	Eq. 12.21				
Stress	Safety Factor	3.01	3.01					
Contact Ratio Check	Base Pitch (pb)	2.8024057						
	Addendum (a)	0.98425						
	Line of Action (Z)	4.217751399						
	Contact Ratio (mp)	1.505046681						

Figure B-1. Gear Stress Calculations - Gears A & B

Gear Stress Calculations - Gears C & D (20:9)				Legend:	Selected Value	Lookup Value	Calculated	Predetermined
DO NOT MODIFY								
Category	Variable	Gear C	Gear D	Notes & equations				
Gear Specs	N - Number of teeth	45	100	Gear Ratio: 9:20				
	F - Face Width (m)	0.4	0.4	Acceptable range (in):				
	m - Mesh Module (m)	0.025	0.025		7.874	< F <	15.748	
	Φ - Pressure Angle (°)	25	25					
	Tp - Pinion Torque (Nm)	76680.72	76680.72					
	ω - Shaft Speed (rad/s)	209.44	94.248					
	d - Diameter (m)	1.125	2.5	d = N*m				
	R - Pitch Radius (in)	22.15	49.21	R = d/2				
	Pd - Diametral Pitch (in ⁻¹)	1.02	1.02	Pd ~ (1/m)				
	Wt - Tangential Force (N)	136321.28	136321.28	Wt = 2(Tp)/(Dp)				
	C - Center Distance (in)	71.36	71.36	C = R1 + R2				
	Xp - Pinion Addendum Coefficient	0.00	0.00	Assume = 0 (full-depth teeth)				
Bending Stress Correction	ρ - Radius of Curvature (in)	8.69	21.46	Eq. 12.22b				
	Vp - Pitch Velocity (m/s)	117.81	117.81	Vp = ω*(Dp/2)				
	J - Bending Geometry Factor	0.35	0.37	Table 12-12: Interpolated				
	Ka - Application Factor	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)				
	Km - Distribution Factor	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m				
	Kv - Dynamic Factor	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)				
	Qv - Quality Index	12	12	Designated as 12, based on Table 12-6				
	Ks - Size Factor	1	1	Assumed gear tooth size has no influence				
	Kb - Backup Ratio	1	1	All solid gears				
	Ki - Idler Factor	1	1	Both non-idling				
	Cp - Elastic Coefficient (MPa ^{1/2})	191	191	Table 12-18: Steel pinion, steel gear				
	I - Surface Geometry Factor	0.127	0.127	Eq. 12.22a				
Surface Stress Correction	Ca	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)				
	Cm	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m				
	Cv	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)				
	Cs	1	1	Assumed gear tooth size has no influence				
	Cf - Surface Finish Factor	1	1	Surface finish to be specified to meet requirement				
	Sfb' - Uncorrect. Bending-Fatigue Strength (Pa)	230000000	230000000	Selected Material: AISI 4140 Nitrieded				
Bending Strength	Kt - Temperature Factor	1	1	Assume T < 250F				
	Kr - Reliability	1	1	Assume 99% Reliability				
	Kl - Lifetime Factor	0.799	0.799	Figure 12-24, Low estimate for ~10 ¹⁰ load cycles				
	Sfb - Corrected Bending-Fatigue Strength (Pa)	183778858.4	183778858.4	Eq. 12.24a				
Surface Strength	Sfc' - Uncorrect. Surface Fatigue Strength (Pa)	1100000000	1100000000	AISI 4140 Nitrieded: 1100-1250MPa (table 12-21)				
	Cl - Lifetime Factor	0.679	0.679	Figure 12-26, Low estimate for ~10 ¹⁰ load cycles				
	Ch - Hardness Difference Factor	1	1	No difference in hardness				
	Ct - Temperature Factor	1	1	Assume T < 250F				
	Cr - Reliability Factor	1	1	Assume 99% Reliability				
	Sfc - Corrected Surface Fatigue (Pa)	747112078.1	747112078.1	Eq. 12.25				
Category	Output Values	Gear A	Gear B	Notes & equations				
Bending Stress	Bending Stress (Pa)	70108089.42	66318462.96					
	Safety Factor	2.62	2.77					
Surface Stress	Surface Stress (Pa)	396380198.07	396380198.07					
	Safety Factor	3.55	3.55					
Contact Ratio Check	Base Pitch (pb)	2.8024057						
	Addendum (a)	0.98425						
	Line of Action (Z)	4.369180119						
	Contact Ratio (mp)	1.559081941						

Figure B-2. Gear Stress Calculations - Gears C & D

Gear Stress Calculations - Gears E & F (1:2)				Legend: Selected Value Lookup Value Calculated Predetermined			
DO NOT MODIFY							
Category	Variable	Gear E	Gear F	Notes & equations			
Gear Specs	N - Number of teeth	25	50	Gear Ratio: 1:2			
	F - Face Width (m)	0.4	0.4	Acceptable range (in): 15.748 < F < 31.496			
	m - Mesh Module (m)	0.05	0.05				
	Φ - Pressure Angle (°)	25	25				
	Tp - Pinion Torque (Nm)	168098.6	168098.6				
	ω - Shaft Speed (rad/s)	94.248	47.124				
	d - Diameter (m)	1.25	2.5	d = N*m			
	R - Pitch Radius (in)	24.61	49.21	R = d/2 *Imperial for Surf. Geo. Calc			
	Pd - Diametral Pitch (in ⁻¹)	0.51	0.51	Pd ~ (1/m) *Imperial for Surf. Geo. Calc			
	Wt - Tangential Force (N)	268957.76	268957.76	Wt = 2(Tp)/(Dp)			
	C - Center Distance (in)	73.82	73.82	C = R1 + R2 *Imperial for Surf. Geo. Calc			
	Xp - Pinion Addendum Coefficient	0.00	0.00	Assume = 0 (full-depth teeth)			
Bending Stress Correction	ρ - Radius of Curvature (in)	8.85	22.35	Eq. 12.22b *Imperial for Surf. Geo. Calc			
	Vp - Pitch Velocity (m/s)	58.905	58.905	Vp = ω*(Dp/2)			
	J - Bending Geometry Factor	0.33	0.35	Table 12-12: Interpolated			
	Ka - Application Factor	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)			
	Km - Distribution Factor	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m			
	Kv - Dynamic Factor	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)			
	Qv - Quality Index	12	12	Designated as 12, based on Table 12-6			
	Ks - Size Factor	1	1	Assumed gear tooth size has no influence			
	Kb - Backup Ratio	1	1	All solid gears			
	Ki - Idler Factor	1	1	Both non-idling			
	Cp - Elastic Coefficient (MPa ^{1/2})	191	191	Table 12-18: Steel pinion, steel gear			
	I - Surface Geometry Factor	0.117	0.117	Eq. 12.22a			
Surface Stress Correction	Ca	1	1	Table 12-17: Uniform shock (Driving machine: Turbine, Driven machine: Generator)			
	Cm	1.8	1.8	Table 12-16: km = 1.8 if 0.25m < F < 0.5m			
	Cv	1	1	For Qv = 12, Kv = 1 (Eq. 12.16si)			
	Cs	1	1	Assumed gear tooth size has no influence			
	Cf - Surface Finish Factor	1	1	Surface finish to be specified to meet requirement			
	Sfb' - Uncorrect. Bending-Fatigue Strength (Pa)	230000000	230000000	Selected Material: AISI 4140 Nitrieded			
Bending Strength	Kt - Temperature Factor	1	1	Assume T < 250F			
	Kr - Reliability	1	1	Assume 99% Reliability			
	Kl - Lifetime Factor	0.799	0.799	Figure 12-24, Low estimate for ~10 ¹⁰ load cycles			
	Sfb - Corrected Bending-Fatigue Strength (Pa)	183778858.4	183778858.4	Eq. 12.24a			
Surface Strength	Sfc' - Uncorrect. Surface Fatigue Strength (Pa)	1100000000	1100000000	AISI 4140 Nitrieded: 1100-1250MPa (table 12-21)			
	Cl - Lifetime Factor	0.679	0.679	Figure 12-26, Low estimate for ~10 ¹⁰ load cycles			
	Ch - Hardness Difference Factor	1	1	No difference in hardness			
	Ct - Temperature Factor	1	1	Assume T < 250F			
	Cr - Reliability Factor	1	1	Assume 99% Reliability			
	Sfc - Corrected Surface Fatigue (Pa)	747112078.1	747112078.1	Eq. 12.25			
Category	Output Values	Gear A	Gear B	Notes & equations			
Bending Stress	Bending Stress (Pa)	73352116.15	69160566.65				
	Safety Factor	2.51	2.66				
Surface Stress	Surface Stress (Pa)	550082311.13	550082311.13				
	Safety Factor	1.84	1.84				
Contact Ratio Check	Base Pitch (pb)	5.6048114					
	Addendum (a)	1.9685					
	Line of Action (Z)	8.359375486					
	Contact Ratio (mp)	1.491464188					

Figure B-3. Gear Stress Calculations - Gears E & F

Appendix C. Shaft Stress Calculation Sheets

Shaft 1 Stress Calculation				Legend:	Selected Value	Lookup Value	Calculated	Predetermined
Category	Variable	Point A	Point B	Notes & equations				
Shaft Parameters	d - Small Diameter (m)	0.432	0.36	Will determine by testing once sheet is done				
	D - Diameter (m)	0.5184	0.432	Will determine by testing once sheet is done				
	D/d - Ratio of Diameters	1.2	1.2					
	S _{ut} - Ultimate Strength (MPa)	1772	1772	Chosen from Table A-10				
	P - Distance Away from Origin (m)	0.135	0.05					
	r - Fillet Radius (m)	0.01	0.01	Will determine by testing once sheet is done				
	sqrt(a) - Neubers Constant	0.036	0.036					
	q - Notch Sensitivity	0.94573868	0.94573868	Dependent on r and neubers				
Nominal Stress Calculations	I - Area Moment of Inertia (m ⁴)	0.00010685	5.153E-05	$\pi \cdot (d/2)^4 / 64$				
	M _{max} - Moment (Nm)	5829.75	2159.16667	Shown in bending moment diagram (using highest torque)				
	$\sigma_{max, nom}$ - Max Stress Due to Bending (MPa)	11.7847067	7.5422123	$M \cdot d / 2I$				
Nominal Torsion Calculation	J - Polar Moment of Inertia (m ⁴)	0.00021371	0.00010306	$\pi \cdot (d/2)^4 / 32$				
	T _{max} - Torque (Nm)	25910	25910					
	$\tau_{max, nom}$ - Max Torsion (MPa)	26.1882372	45.2532738	$T \cdot d / 2J$				
Stress Concentration	K _t - Stress Concentration Factor	2.20638005	2.12042042	Lookup A and b value				
	K _f - Fatigue Stress Concentration Factor	2.14092027	2.05862492	$1 + q \cdot (K_t - 1)$				
	σ_m - Mean Stress Due to Bending (MPa)	25.2301175	15.5341284	$\sigma_{max, nom} \cdot K_f$				
Shear Concentrations	K _{ts} - Stress Concentration Factor (Shear)	1.88522023	1.81225849	Lookup A and b value				
	K _{fs} - Fatigue Stress Concentration Factor (Shear)	1.83718701	1.76818427	$1 + q \cdot (K_{ts} - 1)$				
	τ_m - Mean Torsion (MPa)	48.1126891	80.016127	$\tau_{max, nom} \cdot K_f$				
Stress Correction	S _e - Uncorrected Endurance Limit (MPa)	700	700	If S _{ut} > 1400 MPa S _e = 700 MPa else S _e = 0.5S _{ut}				
	C _{load} - Load Factor	1	1	Bending/Torsion C _{load} = 1				
	C _{size} - Size Factor	0.6	0.6	If d > .25m C _{size} = 0.6, else C _{size} = 1.189d ^{-0.097}				
	C _{surf} - Surface Factor	0.62134621	0.62134621	Machined SF C _{surf} = 4.51(S _{ut}) ^{-0.265}				
	C _{temp} - Temperature Factor	1	1	T < 450 C, C _{temp} = 1				
	C _{reliab} - Reliability Factor	0.814	0.814	99% Reliability, C _{reliab} = 0.814				
	S _e - Corrected Endurance Limit (MPa)	212.425844	212.425844	S _e = $\Pi C_i S_e$				
Safety Factors	σ' - Von Mises Stress	87.0692333	139.459854					
	N _f - Fatigue Safety Factor	2.43973486	1.52320426					

Figure C-1. Shaft 1 Stress Calculations

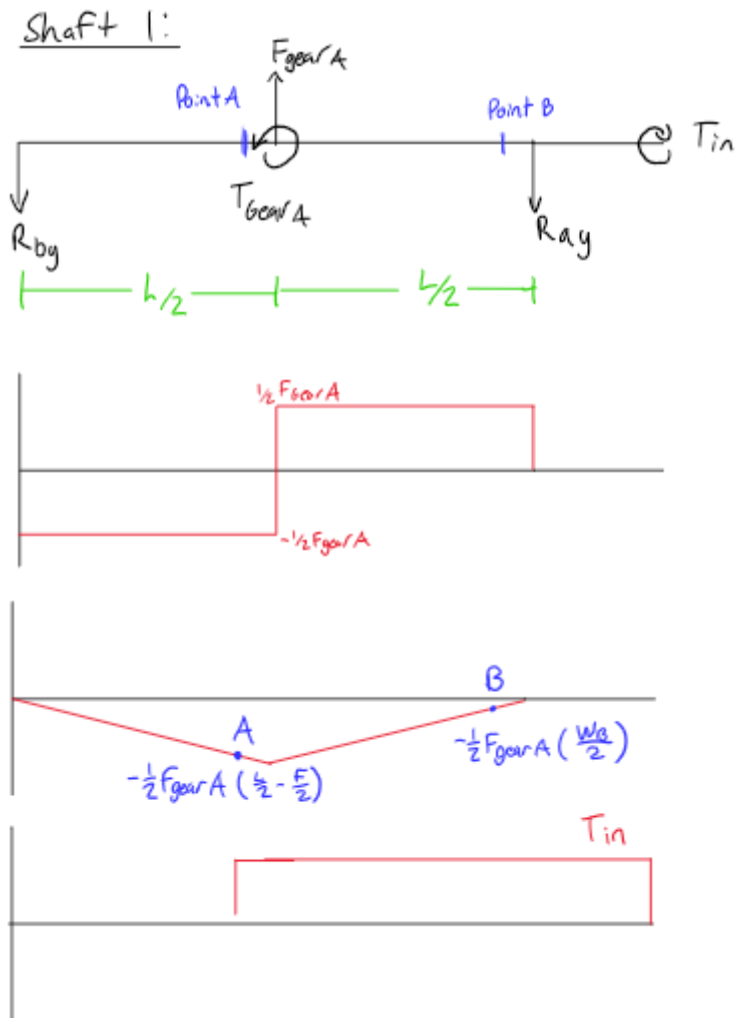


Figure C-2. Shaft 1 Diagrams

Shaft 2 Stress Calculation					Legend: Selected Value	Lookup Value	Calculated	Predetermined
Category	Variable	Point A	Point B	Notes & equations				
Shaft Parameters	d - Small Diameter (m)	0.71	0.6	Will determine by testing once sheet is done				
	D - Diameter (m)	0.85	0.71	Will determine by testing once sheet is done				
	D/d - Ratio of Diameters	1.2	1.2					
	S _{ut} - Ultimate Strength (MPa)	1772	1772	Chosen from Table A-10				
	P - Distance Away from Origin (m)	0.775	0.05					
	r - Fillet Radius (m)	0.01	0.01	Will determine by testing once sheet is done				
	sqrt(a) - Neubers Constant	0.036	0.036					
	q - Notch Sensitivity	0.945738675	0.945738675	Dependent on r and neubers				
Nominal Stress Calculations	I - Area Moment of Inertia (m ⁴)	0.000779621	0.000397608	$\pi*(d/2)^4/64$				
	M _{max} - Moment (Nm)	103287.3889	-6122.25	Shown in bending moment diagram				
	$\sigma_{max, nom}$ - Max Stress Due to Bending (MPa)	47.0318802	-4.61931315	$M*d/2/I$				
Nominal Torsion Calculation	J - Polar Moment of Inertia (m ⁴)	0.001559241	0.000795216	$\pi*(d/2)^4/32$				
	T _{max} - Torque (Nm)	76680.7228	0					
	$\tau_{max, nom}$ - Max Torsion (MPa)	17.45827155	0	$T*d/2/J$				
Stress Concentration	K _t - Stress Concentration Factor	2.45872838	2.370151631	Lookup A and b value				
	K _f - Fatigue Stress Concentration Factor	2.379575846	2.295805389	$1+q*(K_t-1)$				
	σ_m - Mean Stress Due to Bending (MPa)	111.9159261	-10.605044	$\sigma_{max, nom} * K_f$				
Shear Concentrations	K _{ts} - Stress Concentration Factor (Shear)	2.099303043	2.024175572	Lookup A and b value				
	K _{fs} - Fatigue Stress Concentration Factor (Shear)	2.039653403	1.968602449	$1+q*(K_{ts}-1)$				
	τ_m - Mean Torsion (MPa)	35.60882299	0	$\tau_{max, nom} * K_f$				
Stress Correction	S _e - Uncorrected Endurance Limit (MPa)	700	700	If S _{ut} > 1400 MPa S _e = 700 MPa else S _e = 0.5S _{ut}				
	C _{load} - Load Factor	1	2	Bending/Torsion C _{load} = 1				
	C _{size} - Size Factor	0.6	0.6	If d > .25m C _{size} = 0.6, else C _{size} = $1.189d^{0.097}$				
	C _{surf} - Surface Factor	0.621346214	0.621346214	Machined SF C _{surf} = $4.51(S_{ut})^{0.265}$				
	C _{temp} - Temperature Factor	1	1	T < 450 C, C _{temp} = 1				
	C _{reliab} - Reliability Factor	0.814	0.814	99% Reliability, C _{reliab} = 0.814				
	S _e - Corrected Endurance Limit (MPa)	212.4258438	424.8516876	S _e = $\pi C S_e$				
Safety Factors	σ' - Von Mises Stress	127.7855209	10.60504402					
	N _f - Fatigue Safety Factor	1.662362389	40.06128469					

Figure C-3. Shaft 2 Stress Calculations

Shaft 2:

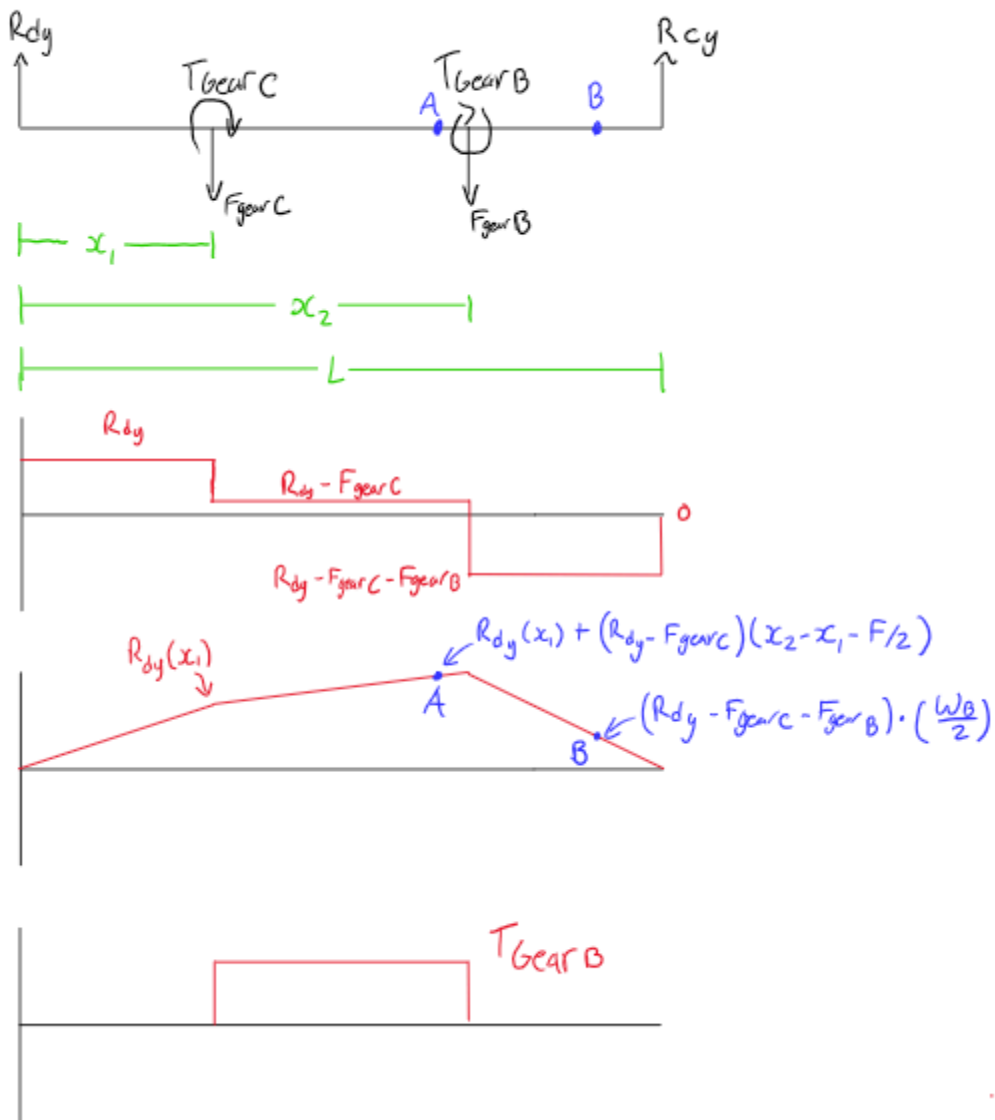


Figure C-4. Shaft 2 Diagrams

Shaft 3 Stress Calculation					Legend: Selected Value	Lookup Value	Calculated	Predetermined
Category	Variable	Point A	Point B	Notes & equations				
Shaft Parameters	d - Small Diameter (m)	0.875	0.75	Will determine by testing once sheet is done				
	D - Diameter (m)	1.05	0.875	Will determine by testing once sheet is done				
	D/d - Ratio of Diameters	1.2	1.2					
	S _u - Ultimate Strength (MPa)	1772	1772	Chosen from Table A-10				
	P - Distance Away from Origin (m)	0.775	0.05					
	r - Fillet Radius (m)	0.01	0.01	Will determine by testing once sheet is done				
	sqrt(a) - Neubers Constant	0.036	0.036					
	q - Notch Sensitivity	0.945738675	0.945738675	Dependent on r and neubers				
Nominal Stress Calculations	I - Area Moment of Inertia (m ⁴)	0.001798383	0.000970722	$\pi * (d/2)^4 / 64$				
	M _{max} - Moment (Nm)	183573.0173	-11605.71	Shown in bending moment diagram				
	σ _{max, nom} - Max Stress Due to Bending (MPa)	44.68856619	-4.48340724	$M * d / 2I$				
Nominal Torsion Calculation	J - Polar Moment of Inertia (m ⁴)	0.003596765	0.001941444	$\pi * (d/2)^4 / 32$				
	T _{max} - Torque (Nm)	168098.5995	0					
	τ _{max, nom} - Max Torsion (MPa)	20.44702033	0	$T * d / 2J$				
Stress Concentration	K _t - Stress Concentration Factor	2.573299643	2.488276497	Lookup A and bvalue				
	K _f - Fatigue Stress Concentration Factor	2.48793032	2.407520643	$1 + q * (K_t - 1)$				
	σ _m - Mean Stress Due to Bending (MPa)	111.1074009	-10.7938955	$\sigma_{max, nom} * K_f$				
Shear Concentrations	K _{ts} - Stress Concentration Factor (Shear)	2.196451099	2.124360552	Lookup A and bvalue				
	K _{fs} - Fatigue Stress Concentration Factor (Shear)	2.131530078	2.063351258	$1 + q * (K_{ts} - 1)$				
	τ _m - Mean Torsion (MPa)	43.58343882	0	$\tau_{max, nom} * K_f$				
Stress Correction	S _e - Uncorrected Endurance Limit (MPa)	700	700	If S _u > 1400 MPa S _e = 700 MPa else S _e = 0.5S _u				
	C _{load} - Load Factor	1	2	Bending/Torsion C _{load} = 1				
	C _{size} - Size Factor	0.6	0.6	If d > .25m C _{size} = 0.6, else C _{size} = 1.189d ^{-0.097}				
	C _{surf} - Surface Factor	0.621346214	0.621346214	Machined SF C _{surf} = 4.51(S _u) ^{-0.265}				
	C _{temp} - Temperature Factor	1	1	T < 450 C, C _{temp} = 1				
	C _{reliab} - Reliability Factor	0.814	0.814	99% Reliability, C _{reliab} = 0.814				
	S _e - Corrected Endurance Limit (MPa)	212.4258438	424.8516876	S _e = ΠC _e S _e				
Safety Factors	σ' - Von Mises Stress	134.3257345	10.79389548					
	N _f - Fatigue Safety Factor	1.581423281	39.36036703					

Figure C-5. Shaft 3 Stress Calculations

Shaft 3:

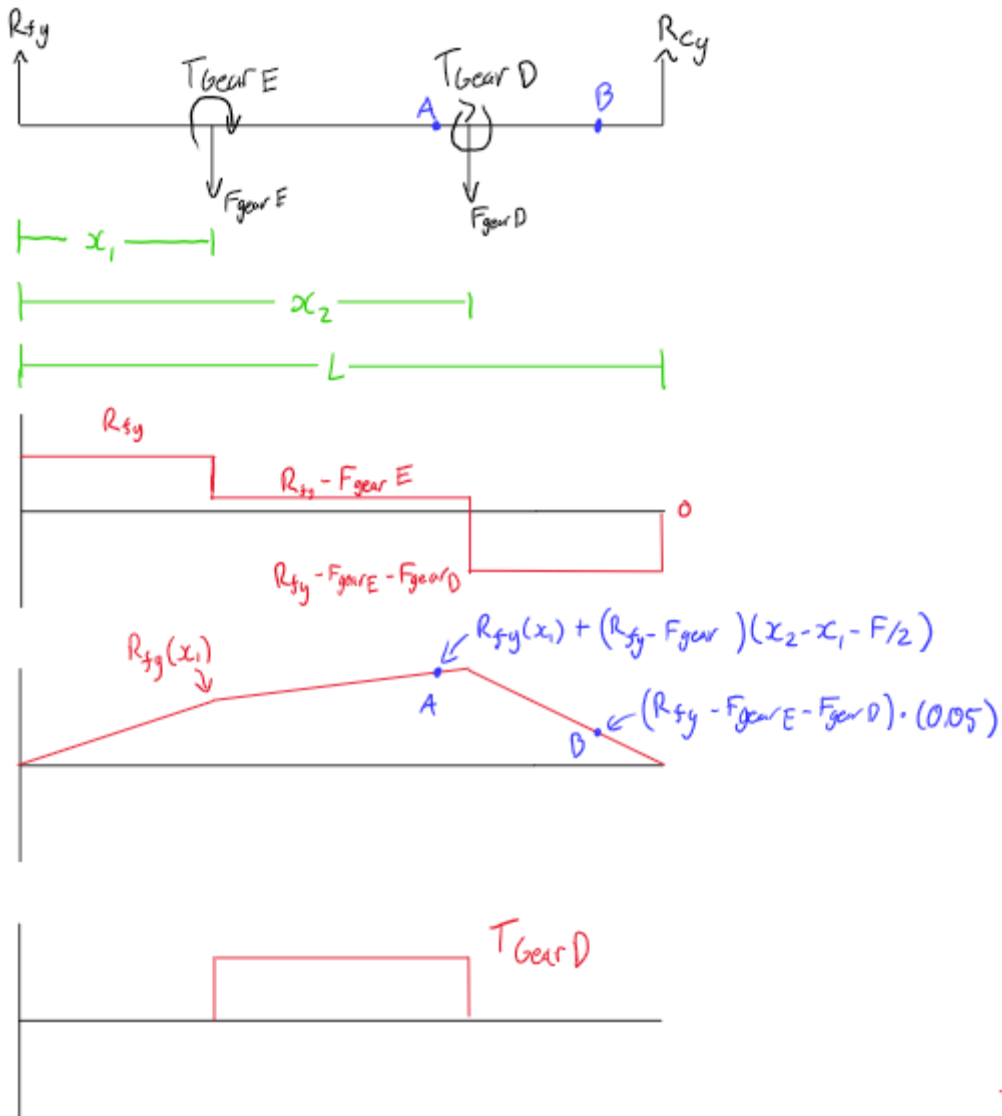


Figure C-6. Shaft 3 Diagrams

Shaft 4 Stress Calculation				Legend:	Selected Value	Lookup Value	Calculated	Predetermined
Category	Variable	Point A	Point B	Notes & equations				
Shaft Parameters	d - Small Diameter (m)	1.0125	0.9	Will determine by testing once sheet is done				
	D - Diameter (m)	1.215	1.215	Will determine by testing once sheet is done				
	D/d - Ratio of Diameters	1.2	1.35					
	S _{ut} - Ultimate Strength (MPa)	1772	1772	Chosen from Table A-10				
	P - Distance Away from Origin (m)	0.3	0.05					
	r - Fillet Radius (m)	0.01	0.01	Will determine by testing once sheet is done				
	sqrt(a) - Neubers Constant	0.036	0.036					
	q - Notch Sensitivity	0.945738675	0.945738675	Dependent on r and neubers				
Nominal Stress Calculations	I - Area Moment of Inertia (m ⁴)	0.00322426	0.00201289	$\pi \cdot (d/2)^4 / 64$				
	M _{max} - Moment (Nm)	40343.66388	6723.94398	Shown in bending moment diagram (using highest torque)				
	$\sigma_{\max, \text{nom}}$ - Max Stress Due to Bending (MPa)	6.334470668	1.503199582	$M \cdot d / I$				
Nominal Torsion Calculation	J - Polar Moment of Inertia (m ⁴)	0.00644852	0.004025779	$\pi \cdot (d/2)^4 / 32$				
	T _{max} - Torque (Nm)	331653.4366	331653.4366					
	$\tau_{\max, \text{nom}}$ - Max Torsion (MPa)	26.03691341	37.0720896	$T \cdot d / J$				
Stress Concentration	K _t - Stress Concentration Factor	2.656477768	2.990630889	Lookup A and b value				
	K _f - Fatigue Stress Concentration Factor	2.56659509	2.88261662	$1 + q \cdot (K_t - 1)$				
	σ_m - Mean Stress Due to Bending (MPa)	16.25802132	4.333148099	$\sigma_{\max, \text{nom}} \cdot K_f$				
Shear Concentrations	K _{ts} - Stress Concentration Factor (Shear)	2.266961715	2.209887555	Lookup A and b value				
	K _{fs} - Fatigue Stress Concentration Factor (Shear)	2.198214694	2.144237454	$1 + q \cdot (K_{ts} - 1)$				
	τ_m - Mean Torsion (MPa)	57.23472566	79.49136302	$\tau_{\max, \text{nom}} \cdot K_f$				
Stress Correction	S _e - Uncorrected Endurance Limit (MPa)	700	700	If S _{ut} > 1400 MPa S _e = 700 MPa else S _e = 0.5S _{ut}				
	C _{load} - Load Factor	1	1	Bending/Torsion C _{load} = 1				
	C _{size} - Size Factor	0.6	0.6	If d > .25m C _{size} = 0.6, else C _{size} = $1.189d^{-0.097}$				
	C _{surf} - Surface Factor	0.621346214	0.621346214	Machined SF C _{surf} = $4.51(S_{ut})^{-0.265}$				
	C _{temp} - Temperature Factor	1	1	T < 450 C, C _{temp} = 1				
	C _{reliab} - Reliability Factor	0.814	0.814	99% Reliability, C _{reliab} = 0.814				
	S _e - Corrected Endurance Limit (MPa)	212.4258438	212.4258438	$S_e = \text{PCS}_e$				
Safety Factors	σ' - Von Mises Stress	100.4577758	137.7512488					
	N _f - Fatigue Safety Factor	2.114578409	1.542097408					

Figure C-7. Shaft 4 Stress Calculations

Shaft 4:

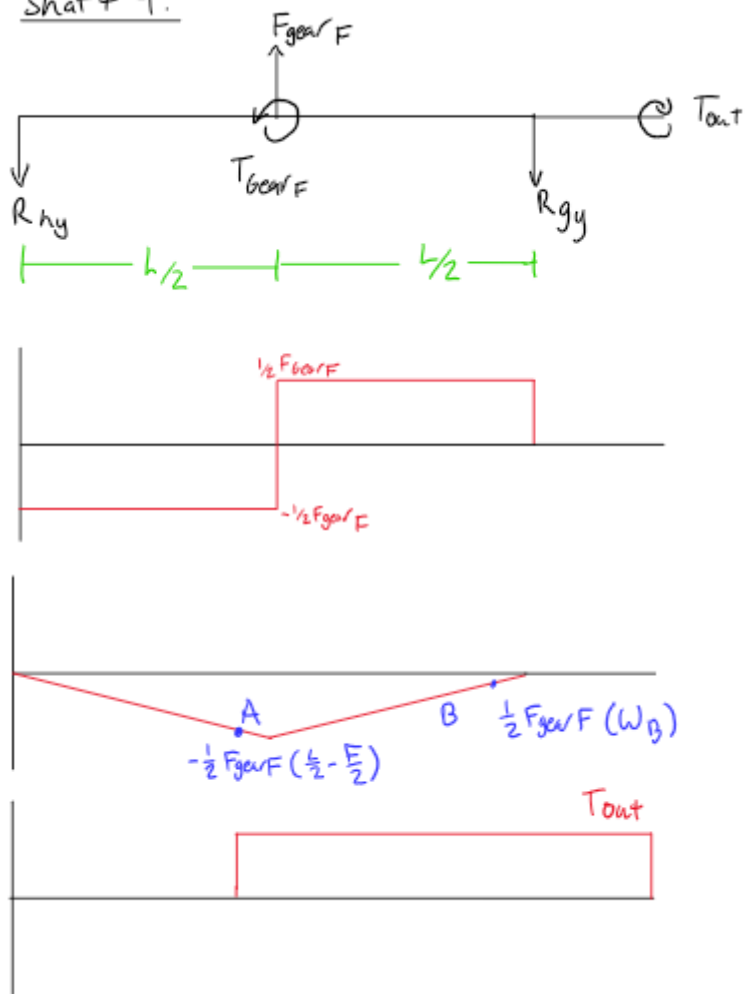


Figure C-8. Shaft 4 Diagrams